



L^AT_EX M1 NOTE II

X
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Github: <https://github.com/Baudelaireee/Notebook>

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1 Some group and linear algebra

This section can be seen as a brief review of linear algebra and group theory, it mainly focus on something related to lie theory and representation theory.

1.1 Decomposition of endomorphisms

Theorem 1.1 (Real-Polar decomposition).

For any real endomorphism $A \in \text{GL}_n(\mathbb{R})$, there exists a unique orthogonal endomorphism $O \in O(n)$ and a unique symmetric positive definite endomorphism $P \in S_n^{++}(\mathbb{R})$ such that

$$A = OP$$

Proof. Here is the outline of the proof:

- (1) A **lemma** about the existence of square root are needed here: If $A \in S_n^+(\mathbb{R})$, then there exists a (unique) $B \in S_n^+(\mathbb{R})$ such that $B^2 = A$.
- (2) Prove that $A^t A$ is (semi-definite) positive, thus by the lemma there exist a (unique) square root $P = \sqrt{A^t A}$, and we prove that P is positive definite.
- (3) Define $O = AP^{-1}$, and prove that O is orthogonal.
- (4) Prove the uniqueness of the decomposition.

Here is the details:

- (1) By real-spectral theorem, there exists $O \in O(n)$ such that

$$A = ODO^T, \quad D = \text{dig}(\lambda_1, \dots, \lambda_n)$$

where $\lambda_i \geq 0$ are the eigenvalues of A , and we set $\Sigma = \text{dig}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})$ such that

$$A = (O\Sigma O^T)^2$$

which shows that $B = O\Sigma O^T$ is a (unique) square root of A and it is positive since $\sqrt{\lambda_i} \geq 0$.

- (2) $A^t A$ is clearly symmetric, and it is positive since

$$x^T (A^t A) x = \|Ax\|^2 \geq 0$$

for any $x \in \mathbb{R}^n$ under the common inner product, and A is invertible such that $\|Ax\| = 0$ if and only if $x = 0$, thus $A^t A$ is positive definite, so does its square root (the eigenvalues is strictly positive).

- (3) $P^{-1} = (P^{-1})^T$ since P is symmetric, so we have

$$O^T O = (P^{-1})^T A^t A P^{-1} = (P^{-1})^T P^2 P^{-1} = I$$

- (4) Suppose that there exists another decomposition $A = O_0 P_0$, then we have

$$A^t A = P_0^T O_0^T O_0 P_0 = P_0^2$$

so P_0 is the positive square root of $A^t A$, if the uniqueness of (1) is proved, we can finish the proof, but here we give another proof of the uniqueness only under the existence of the

square root: we have $O_0^{-1}O = P_0P^{-1}$, so it is easy to check that P_0P^{-1} is **orthogonal**; notice that P_0 and $A^T A$ commute, and there exists a polynomial f such that

$$f(A^T A) = P$$

it can be done by using lagrange interpolation, thus P_0 and P commute as two symmetric matrix, then P and P_0 can be diagonalized simultaneously by one orthogonal H such that

$$P = HDH^T \quad P_0 = HD_0H^T$$

where $D = \text{dig}(\lambda_1, \dots, \lambda_n)$ and $D_0 = \text{dig}(\mu_1, \dots, \mu_n)$, thus

$$P_0P^{-1} = HD_0^{-1}DH^T$$

it is **symmetric and positive definite** since $D_0^{-1}D$ is digonal with positive entries, thus $P_0P^{-1} \in O(n) \cap S_n^{++}(\mathbb{R}) = \{I\}$, which implies that $P_0 = P$ and $O_0 = O$. $\square$

Theorem 1.2 (Complex-Polar decomposition).

For any complex endomorphism $A \in \text{GL}_n(\mathbb{C})$, there exists a unique unitary endomorphism $U \in U(n)$ and a unique hermitian positive definite endomorphism $P \in H_n^{++}(\mathbb{C})$ such that

$$A = UP$$

The proof is similar to the real case, we set $P = \sqrt{A^*A}$ and $U = AP^{-1}$, and the lemma is based on the complex-spectral theorem for normal operator: The postive Hermitian endomorphism has a unique positive Hermitian square root.

Remark 1.1. In particular, if $n = 1$, we have the polar decompistion in $\text{GL}_1(\mathbb{C}) = \mathbb{C}^*$ by

$$z = |z|e^{i\text{Arg}(z)}$$

it is easy to see that $U(1) \cong \mathbb{S}^1$, it is the original ideal of the decomposition. Similarly, in the real case, we can embed $\mathbb{C}$ into $\text{GL}_2(\mathbb{R})$ by

$$a + ib \mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

and we have the polar decomposition for the matrix form of complex numbers:

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} \sqrt{a^2 + b^2} & 0 \\ 0 & \sqrt{a^2 + b^2} \end{pmatrix} \begin{pmatrix} \frac{a}{\sqrt{a^2 + b^2}} & -\frac{b}{\sqrt{a^2 + b^2}} \\ \frac{b}{\sqrt{a^2 + b^2}} & \frac{a}{\sqrt{a^2 + b^2}} \end{pmatrix}$$

Another useful decomposition is QR decomoposition, which is also called as Gram-Schimdt decompoistion. Suppose that

2 Topological Groups

2.1 Basic properties

Group and Topological space is two different objects, group is algebraic structure, topological space is geometric structure, but it is not strange to combine them together.

Definition 2.1. A **topological group** is a object $(G, \mathcal{T}, \cdot)$ such that

- $(G, \cdot)$ is a group
- $(G, \mathcal{T})$ is a topological space
- The group structure is compatible with topological structure, i.e. multiplication

$$m : G \times G \rightarrow G, \quad (x, y) \mapsto x \cdot y$$

and inverse

$$i : G \rightarrow G, \quad x \mapsto x^{-1}$$

are continuous maps.

With the basic definition, we can talk about the properties of topological groups, here is a useful lemma, it shows that the topological group is **homogeneous**, i.e. the local topology is same at any point.

Lemma 2.1. Let G be a topological group, then for any $g \in G$, we define **left translation** and **right translation** by

$$\begin{aligned} L_g : G &\rightarrow G, & x &\mapsto gx \\ R_g : G &\rightarrow G, & x &\mapsto xg \end{aligned}$$

Then both L_g and R_g are homeomorphisms on G .

Proof. It is easy to verify that $L_{gh} = L_g \circ L_h$, and $R_{gh} = R_h \circ R_g$, then we have

$$L_g^{-1} = L_{g^{-1}}, \quad R_g^{-1} = R_{g^{-1}}$$

and we can check the continuity of L_g and R_g by composition. □

It is an important in the following proofs of properties, we first talk about the basic properties of topological groups.

Proposition 2.2. Let G be a topological group, then

- (1) each open subgroup $H < G$ is also closed.
- (2) each closed subgroup $H < G$ with finite index is also open.

Proof. The subgroup H gives a partition of G by left cosets:

$$G = \bigsqcup_{g \in \mathcal{S}} gH$$

with $\mathcal{S} \subset G$ the set of representatives in G/H , and we fix the representatives for H be e , then

$$G - H = \bigsqcup_{g \in \mathcal{S} - \{e\}} gH$$

If H is open, then by translation each gH is also open, thus the union of open sets $G - H$ is also open, hence H is closed, which finishes that proof of (1). For the statement (2), the finite index shows that $\mathcal{S}$ is finite, and we suppose that the index is n . If $n = 1$, then $H = G$ is open clearly; otherwise, we can deduce that $G - H$ is the finite union of closed sets gH with $g \in \mathcal{S} - \{e\}$, thus it is closed, it can be done similarly by translation, hence H is open since $G - H$ is closed. $\square$

Then it is the **separation** of topological groups.

Proposition 2.3. Let G be a topological groups, then the following statements are equivalent:

- (1) G is Hausdorff (T2).
- (2) The singleton $\{e\}$ is closed in G .
- (3) G is T1.

Proof. (3) implies (2) is clear, since T1 space makes all singletons closed; (1) implies (3) is just the implication from strong to weak separation axiom.

(2) implies (1): G is Hausdorff if and only if the diagonal set

$$\Delta = \{(g, g) \in G \times G \mid g \in G\}$$

is closed under the product topology, and topolglcal group induces a continous map $f(x, y) = xy^{-1}$ for any $(x, y) \in G \times G$, then $\Delta = f^{-1}(\{e\})$ is closed by continuity. $\square$

Corollary 2.4. Let G be a topological group and $H < G$ be a subgroup, then the quotient space G/H is **Hausdorff** if and only if H is **closed** in G .

Another topological properties is about **connectness**:

Proposition 2.5. Let G be a topological group, then

- (1) If G is connected, then any open set containing e generates G .
- (2) The connected component G_0 of identity $e \in G$ is a **closed normal subgroup** of G , and each connected component of G is homeomorphic to G_0 by translation.

Proof. (1) We denote $\langle A \rangle$ be the subgroup generated by $A \subset G$, and we take A be an open set containing e , we will prove that $\langle A \rangle$ is open and closed, by the connectness of G , which implies that $\langle A \rangle = G$. It is clear that

$$\bigcup_{x \in \langle A \rangle} xA \subset \langle A \rangle$$

conversely, for any $a \in \langle A \rangle$, we have $a = ae$ with $e \in A$, so a is in the union such that we can replace the inclusion by equality. Notice that $xA = L_x(A)$ is open since A is open, so $\langle A \rangle$ is also open as the union of open sets; By the proposition 2.2, $\langle A \rangle$ is also closed since it is open subgroup, hence we finish the proof of (1).

(2) Any connected component is closed in general topology, which can be proved by contradiction to $G_0 \subsetneq \overline{G_0}$. To prove that G_0 is a subgroup, we take a continuous map $f(x, y) = xy^{-1}$ on $G \times G$, then $f(G_0 \times G_0)$ is connected since the finite product of connected spaces is connected, and we notice that $a = f(a, e)$ for any $a \in G_0$, so $G_0 \subset f(G_0 \times G_0)$ and then we can take equality by the maximality of connected component, and it shows that G_0 is exactly a subgroup.

To prove the normality, we take the conjugation map for any $g \in G$:

$$\sigma_g : G \rightarrow G, \quad x \mapsto gxg^{-1}$$

it is homeomorphism by composition $\sigma_g = L_g \circ R_g^{-1}$. Here $\sigma_g(e) = e$ and the image is also connected, so $\sigma_g(G_0) \subset G_0$ by the maximality of connected component, so it must be normal. $\square$

Notice that if $H < G$ is a subgroup, then it induces a natural equivalence relation on G by left cosets:

$$x \sim y \iff x = yh, \text{ for some } h \in H$$

similarly we can define right cosets here, and then it gives a quotient space $G/\sim$ and we denote it by G/H , and it is naturally endowed with the quotient topology with a (continuous) quotient map:

$$\pi : G \rightarrow G/H, \quad g \mapsto gH$$

In particular, if H is a normal subgroup, then G/H is exactly a topological group by UPQ:

$$\begin{array}{ccc} G \times G & \xrightarrow{f} & G \\ \pi \times \pi \downarrow & & \downarrow \pi \\ G/N \times G/N & \xrightarrow{f_{G/N}} & G/N \end{array}$$

with $f(g, h) = gh^{-1}$, so it gives the continuity of group operation. However, it is not so easy to talk about the topology of quotient space G/H directly, it is better to see the quotient space as a **homogeneous space** under the action of G , hence we need to talk about action of the topological group:

Definition 2.2. Let G be a topological group and X be a topological space, a continuous action of G on X is a well-defined action in the sense of group together with a continuous operation, that means there exists a continuous map

$$\phi : G \times X \rightarrow X, \quad (g, x) \mapsto g \cdot x$$

such that for any $g, h \in G$ and $x \in X$, we have

$$(1) \quad e \cdot x = x$$

$$(2) \quad g \cdot (h \cdot x) = (gh) \cdot x$$

In the context of topological groups or Lie groups, we always omit "continuous" when it comes to actions, it is a convention sometimes in the literature. It is similar to the algebraic action, we can define **stabilizer** of a point $x \in G$

$$I_x = \{g \in G \mid g \cdot x = x\}$$

It is sometimes called **isotropy group** at x , and it can be written as the preimage

$$I_x = \phi_x^{-1}(\{x\}), \quad \phi_x(g) = g \cdot x$$

with ϕ_x a continuous map by fixing variable, hence the isotropy group is always a **closed subgroup** of G if X is Hausdorff (T1 is enough, but manifold makes sense). Similarly, we can define **orbit** of x by

$$Gx = O_x = \{g \cdot x \mid g \in G\}$$

which is the image of ϕ_x on the contrary, hence it gives a natural bijection

Lemma 2.6 (stabilizer-orbit).

Let G be a compact group and X be a Hausdorff space, If G acts continuously on X , then for any $x \in G$, there exists a homeomorphism $G/I_x \cong Gx$.

Proof. The proof is just based on the universal property of quotient space, and the stabilizer-orbit theorem in group theory, we just need to check the natural bijection induced by ϕ_x is indeed a homeomorphism. $\square$

In particular, the orbit gives an equivalence relation on X by

$$x \sim y \iff y = g \cdot x, \text{ for some } g \in G$$

hence it gives a quotient space denoted by X/G and we call it **orbit space**, it is naturally endowed with the quotient topology with a (continuous) quotient map:

$$\pi : X \rightarrow X/G, x \mapsto Gx$$

the following lemma is useful and important, it shows that the topology of group action is well-behaved from the view of technique:

Lemma 2.7. The natural quotient map $\pi : X \rightarrow G/X$ is open.

Proof. Suppose that U is an open set of X , then we can prove

$$\pi^{-1}(\pi(U)) = \bigsqcup_{g \in G} gU$$

and then by translation each gU is open, so we can finish the proof by the definition of quotient topology that $\pi(U)$ is open if its preimage is open. $\square$

Then we can conclude

Proposition 2.8. Let G be a topological group and H be a subgroup, then

- (1) H is closed if and only if G/H is Hausdorff.
- (2) If G is connected, then G/H is connected.
- (3) If G/H and H are both connected, then G is connected.

Proof. □

Neighborhood system

As we have mentioned before, topological group is a homogeneous space, so the local topology at one point can be translated to generate the global topology, for some convenience, we always study the local topology at identity of the group.

Formally, for any group G we define the **neighborhood system** (at x) to be the collection of subsets containing x , denoted by $\mathcal{V}_x$:

$$A \in \mathcal{V}_x \iff x \in A \wedge A \subset G$$

Hence we can induce a topology on G by neighborhood system (i.e. determine the open voisinage of identity).

Proposition 2.9. Let G be a group with $\mathcal{V}_e$ as the neighborhood system at identity e , if $\mathcal{V}_e$ satisfies the following conditions:

1. For any $U, V \in \mathcal{V}_e$, there exists $W \in \mathcal{V}_e$ such that

$$W \subset U \cap V.$$

2. If $a \in U \in \mathcal{V}_e$, then there exists $V \in \mathcal{V}_e$ such that

$$Va \subset U.$$

3. For each $U \in \mathcal{V}_e$, there exists $V \in \mathcal{V}_e$ such that

$$V^{-1}V \subset U.$$

4. For each $U \in \mathcal{V}_e$ and each $x \in G$, there exists $V \in \mathcal{V}_e$ such that

$$x^{-1}Vx \subset U.$$

then there exist a unique topology $\mathcal{T}$ such that $\mathcal{V}_e$ is the set of all open voisinage of e , and $(G, \mathcal{T})$ is a topological group.

Proof. The topology $\mathcal{T}$ is uniquely defined by translation:

$$\mathcal{B} = \{gU \mid g \in G, U \in \mathcal{V}_e\}$$

we just need to prove that $\mathcal{B}$ is the topology basis. Notice the condition (1) and (2) ensure that $\mathcal{B}$ generates a topology, the condition (3) is the requirement of continuity of multiplication and inverse; finally, the condition (4) is obligator for non-abelian group to ensure the continuity. □

In particular, we can naturally choose subgroup as the element in the system, the condition can be simplified as following:

Corollary 2.10. Let G be a group with $\mathcal{U}$ as a collection of subgroups of G , if $\mathcal{U}$ satisfies the following conditions:

1. For any $U, V \in \mathcal{U}$, there exists $W \in \mathcal{U}$ such that

$$W \subset U \cap V.$$

2. For each $U \in \mathcal{U}$ and each $x \in G$, there exists $V \in \mathcal{U}$ such that

$$x^{-1}Vx \subset U.$$

then there exist a unique topology $\mathcal{T}$ such that $\mathcal{U}$ is a basis of open voisinage of e , and $(G, \mathcal{T})$ is a topological group.

Here are some classic examples of topological groups constructed by neighborhood system:

Example 2.1. We consider the usual topology on $\mathbb{Q}$, then the completion of $\mathbb{Q}$ is just $\mathbb{R}$, it is another method to construct real number (see in Bourkabi's book). We review that we always write the voisinage of 0 as the form of $(-\varepsilon, \varepsilon)$, which motivates us to consider the system of subgroup:

$$\mathcal{U} = \left\{ \frac{1}{n}\mathbb{Z} \mid n \in \mathbb{N}^* \right\}$$

pay attention to the choices here! We can not take $n\mathbb{Z}$ to be an open set (consider it in $\mathbb{R}$, it is closed and not open). Then we can verify the topology induced by $\mathcal{U}$ is just the subspace topology induced by $\mathbb{R}$, i.e. it induces a usual absolute value on field $\mathbb{Q}$:

$$|x|_\infty = \begin{cases} x & x \geq 0 \\ -x & x \leq 0 \end{cases}$$

It is natural to ask that can we define other topological structure on $\mathbb{Q}$ such that we can get other different completion? The answer is yes, and a classic example is the *p -adic number field*.

Example 2.2. We fix a prime number p and then we can define a subgroup

$$\mathbb{Z}_{(p)} = \left\{ \frac{m}{n} \mid m, n \in \mathbb{Z}, p \nmid n \right\}$$

and we let $\mathcal{U} = \{p^t\mathbb{Z}_{(p)} \mid t \in \mathbb{Z}\}$ to be the collection of subgroups, which generates an unique topology (verify!). Notice that we have a unique chain in the system:

$$\dots p^2\mathbb{Z}_{(p)} \subset p\mathbb{Z}_{(p)} \subset \mathbb{Z}_{(p)} \subset p^{-1}\mathbb{Z}_{(p)} \subset p^{-2}\mathbb{Z}_{(p)} \subset \dots$$

motivates from the above example, we consider each subgroup gives a method to measure the size of an element in some sense, for example we consider the following number when $p = 3$:

$$1, \quad 4, \quad \frac{1}{2}, \quad \frac{2}{5}$$

they are all in $\mathbb{Z}_{(p)}$, and we consider the distance of them to 0 can be bounded by α (consider $(-\varepsilon, \varepsilon)$ gives the set of all elements with distance to 0 less than ε), and then we consider some other numbers:

$$\frac{1}{3}, \quad \frac{2}{3}, \quad \frac{7}{6}$$

they are all in $\frac{1}{3}\mathbb{Z}_{(p)}$ but not in $\mathbb{Z}_{(3)}$, so we can think that their distance to 0 is bounded by β , and it is natural to give a relation $\alpha < \beta$, which motivates us to give a new method to consider the metric on $\mathbb{Q}$ by consider the local factorization of p .

Hence we can define something to denote α and β just now:

$$v_p : \mathbb{Z} - \{0\} \rightarrow \mathbb{N}, \quad m \mapsto \max\{k : p^k | m\}$$

for example $v_3(6) = 2$, $v_3(-2) = 0$. Then the method of measuring can be extended to $\mathbb{Q}$ as following:

$$v_p : \mathbb{Z} - \{0\} \rightarrow \mathbb{Z}, \quad \frac{m}{n} \mapsto v_p(m) - v_p(n)$$

And we make a convention that $v_p(0) = +\infty$, it is called **p-adic valustion**. for example $v_3(\frac{1}{3}) = -1$, $v_3(\frac{9}{2}) = 2$. Then we can get a measure of number

$$p^k \mathbb{Z}_{(p)} \iff \{x \in \mathbb{Q} \mid v_p(x) \geq p^k\}$$

but a question occurs here is that $v_3(1) = 0$ and $v_3(\frac{1}{3}) = -1$, as what we discussed just now, actually we hope the measure of 1 should be less than the measure of $\frac{1}{3}$ (here $0 > -1$), hence the valuation v_p can not be treated as the absolute value directly, hence we modify it as following:

$$|-|_p : \mathbb{Q} \rightarrow \mathbb{R}_+, \quad x \mapsto \frac{1}{e^{v_p(x)}}$$

the convention just now forces $|0|_p = 0$, and again we can get a clear measure of numbers:

$$x \in p^k \mathbb{Z}_{(p)} \iff |x|_p \leq e^{-k}$$

hence $|1|_3 < |\frac{1}{3}|_3$, which admits our original idea.

Remark 2.1. it is just a short introduction of p-adic number from the view of topological group, and here are something to supplement if to go deeper:

(1) The metric of is induced by the absolute value:

$$d(x, y) = |x - y|_p$$

we can verify that is a **ultrametric (non-archimedean metric)**, i.e. a metric satifying the strong triangle inequality:

$$d(x, z) \leq \max\{d(x, y), d(y, z)\}, \quad \forall x, y, z \in \mathbb{Q}$$

which induces different analysis structure on $\mathbb{Q}$.

(2) Same situation with $\mathbb{Q}$ equipped by the usual absolute value, $(\mathbb{Q}, |-|_p)$ is not complete as a metric space, we can complete it to get the **p-adic number field** $\mathbb{Q}_p$.

(The completion process is same with what we do for the real number, and we can prove that the completion is unique up to isometry.)

(3) By **Ostrowski's theorem**, any non-trivial absolute value on $\mathbb{Q}$ is equivalent to either the usual absolute value or some p -adic absolute value. In detail, if $|\cdot|$ is a non-trivial absolute value on $\mathbb{Q}$, then either there is some $c > 0$ such that

$$|x| = |x|_{\infty}^c, \quad \forall x \in \mathbb{Q}$$

or there is some prime p and some $c > 0$ such that

$$|x| = |x|_p^c, \quad \forall x \in \mathbb{Q}$$

2.2 Homogeneous Spaces

Review that a group action $G \curvearrowright X$ is **transitive** if for any $x, y \in X$, there exists $g \in G$ such that $g \cdot x = y$, i.e. there is only one orbit.

Definition 2.3. A **homogeneous space** is a Hausdorff topological space X with a transitive continuous action of a group G , in particular, we call it **G-homogeneous space** to emphasize the acting group G .

Before talking about the properties of homogeneous space, we consider a basic example but important example: rotation action on sphere $SO(n) \curvearrowright \mathbb{S}^{n-1}$

$$SO(n) \times \mathbb{S}^{n-1}, \quad (A, v) \mapsto Av$$

the operation is defined by matrix multiplication, so it defines a continuous action, then we can verify the following results:

1. **The action is transitive.** When $n = 2$, it is clear since $SO(2)$ is just the set of rotations of plane, so we prove the case $n \geq 3$ by induction. We assume that the action is transitive for $n - 1, \dots$
2. **The isotropy group at e_1 is isomorphic to $SO(n-1)$.** Suppose that $A \in SO(n)$ is matrix such that $Ae_1 = e_1$, then by calculation we can verify that A is of the form

$$M = \begin{pmatrix} 1 & 0_{1 \times n} \\ 0_{n \times 1} & A' \end{pmatrix}, \quad A' \in SO(n-1).$$

Then it is easy to establish an isomorphism and a homeomorphism.

3. **There exists a homeomorphism $SO(n)/SO(n-1) \cong \mathbb{S}^{n-1}$.** By the stabilizer-orbit lemma, we can deduce a continuous bijection $SO(n)/SO(n-1) \rightarrow \mathbb{S}^{n-1}$, so it is enough to verify that $SO(n)$ is compact.
4. **$SO(n)$ is connected.** When $n = 2$, a classic homeomorphism $SO(2) \cong \mathbb{S}^1$ shows that $SO(2)$ is connected, then we can prove the case $n \geq 3$ by induction.

From this example, we can see here sphere $\mathbb{S}^n$ is a homogeneous space under the action of $SO(n+1)$, we are very familiar with the topology of sphere, but we are not familiar with the topology of $SO(n)$, so homogeneous space gives us a method to understand the topology of group by the homogeneous space. Conversely, if we are familiar with the structure of the topological group, we can understand a strange space by this method, any homogeneous space gives a deep relation

$$\{\text{topological group}\} \sim \{\text{space with same local structure}\}$$

Theorem 2.1. Let G be a Lie group and X be a G -homogeneous space, if I_x the isotropy group at $x \in X$, then there exists a homeomorphism

$$G/I_x \cong X$$

2.3 The Pontryagin Duality

We consider the category of locally compact abelian groups:

$$\mathbf{LCA} \left| \begin{array}{l} \text{objects: locally compact abelian groups } (G, +) \\ \text{morphisms: continuous group homomorphisms} \end{array} \right.$$

without talking about the topology of the group, we can similarly consider the pure algebraic structure:

$$\mathbf{Ab} \left| \begin{array}{l} \text{objects: abelian group } (G, +) \\ \text{morphisms: group homomorphisms} \end{array} \right.$$

Clearly it is same as category of $\mathbb{Z}$ -modules, hence it is an abelian category with tools of homological algebra, so we can choose a certain objects of abelian groups to construct a duality.

$$\hat{G} := \text{Hom}_{\mathbb{Z}}(G, \mathbb{R}/\mathbb{Z})$$

Clearly, $\mathbb{R}/\mathbb{Z}$ is an injective $\mathbb{Z}$ -module, hence the Hom-functor will keep exactness. In particular, it will have a good behavior when group is finite.

Proposition 2.11. If G is a finite abelian group, then $\hat{G}$ is isomorphic to G .

Proof. □

However, the duality fails when G is infinite. For example, if $G = \mathbb{Z}$, then $\hat{\mathbb{Z}} \cong \mathbb{R}/\mathbb{Z}$, but if we take dual again

$$\hat{\hat{\mathbb{Z}}} = \text{End}_{\mathbb{Z}}(\mathbb{R}/\mathbb{Z}) \cong \prod_p \mathbb{Z}_p$$

clearly it is larger than $\mathbb{Z}$, hence it actually fails to be a duality. What we have done is to give a exact restriction on the category of abelian groups to make the duality work: It depends on the topology of the group, so we need to consider the topology of the dual group firstly. Formally, we define dual group as following:

Definition 2.4. Let $G \in \mathbf{LCA}$, we define its **dual group** as G^* or $\hat{G}$ by

$$G^* := \text{Hom}_{\mathbf{LCA}}(G, \mathbb{S}^1)$$

i.e. the group of continuous group homomorphism from G to the circle group ($\mathbb{S}^1 \cong \mathbb{R}/\mathbb{Z}$). Among G^* , the elements are called **characters** of G , and we usually denote it by

$$\chi : G \rightarrow \mathbb{S}^1$$

with operation defined by pointwise multiplication:

$$(\chi_1 + \chi_2)(g) := \chi_1(g) \cdot \chi_2(g)$$

Generally a Hom-functor is not necessary to keep category itself, it will map to a category of sets, so we should verify that LCA is a good condition to make the functor closed.

Lemma 2.12. Let $G \in \mathbf{LCA}$, then its dual group G^* endowed with the compact-open topology is a locally compact abelian group, in particular

$$V(K, \varepsilon) = \{\chi \in G^* : \chi(K) \subset U_\varepsilon\}$$

for any compact subset $K \subset G$ and $U_\varepsilon = \{e^{it} \mid t \in (-\varepsilon, \varepsilon)\}$, it forms a neighborhood basis of $0 \in G^*$.

Proof. It is easy to verify that G^* is indeed a abelian group under defined operation, to verify it is a topological group, we need to verify the family $\mathcal{B}$ of all $V(K, \text{varepsilon})$ forms a neighborhood basis of identity of G^* : (1) zero character $g \mapsto 1$ is in any set clearly; (2) $V(K, r) \cap V(L, t) = V(K \cup L, \min(r, t))$, so it is stable under finite intersection; (3) $-V(K, \varepsilon) = V(K, \varepsilon)$ since $U_\varepsilon^{-1} = U_\varepsilon$; (4) $V(K, r) + V(L, t)$ is the subset of $V(K \cup L, r + t)$, so it is stable under operation. Hence it forms a neighborhood system of identity by (1)-(4).

Finally, we need to verify that G^* is locally compact by Ascoli's theorem... □

Remark 2.2. In particular, we denote evaluation map by

$$\text{ev} : G \times G^* \rightarrow \mathbb{S}^1, \quad (g, \chi) \mapsto \chi(g)$$

the compact-open topology is **the coarsest topology** such that the evaluation map is continuous (page 43 [1]).

A convention is that G^* always means the dual group with compact-open topology, so we finish the level of objects in the category, and then it is naturally to consider the functoriality, or the level of morphisms.

Lemma 2.13. Let $G, H \in \mathbf{LCA}$, and $f : G \rightarrow H$ be a continuous group homomorphism, then it induces a natural map

$$f^* : H^* \rightarrow G^*, \quad \chi \mapsto \chi \circ f$$

which is a continuous (under the compact-open topology) group homomorphism.

Proof. For any two characters $\chi_1, \chi_2 \in H^*$, we have

$$f^*(\chi_1 + \chi_2)(g) = (\chi_1 + \chi_2)(f(g)) = \chi_1(f(g)) \cdot \chi_2(f(g)) = f^*(\chi_1)(g) \cdot f^*(\chi_2)(g)$$

for any $g \in G$, so $f^*(\chi_1 + \chi_2) = f^*(\chi_1) + f^*(\chi_2)$ it is a group homomorphism. To prove the continuity, we take a basic open set $V_G(K, \varepsilon)$, then

$$\begin{aligned} (f^*)^{-1}(V_G(K, \varepsilon)) &= \{\chi \in H^* \mid \chi \circ f \in V_G(K, \varepsilon)\} \\ &= \{\chi \in H^* \mid \chi(f(K)) \subset U_\varepsilon\} \\ &= V_H(f(K), \varepsilon) \end{aligned}$$

here $f(K)$ is compact since f is continuous, so the preimage of any neighborhood of identity is open, hence f^* is continuous by translation. □

It is easy to verify that $(f \circ g)^* = f^* \circ g^*$, hence we have constructed a contravariant functor inside the category $\mathbf{LCA}$.

Lemma 2.14. If $f : G \rightarrow H$ is a surjective continuous group homomorphism, then $f^* : H^* \rightarrow G^*$ is injective; If $f : G \rightarrow H$ is both injective and open, then $f^* : H^* \rightarrow G^*$ is surjective.

Proof. If f is surjective, we take any $\chi_1, \chi_2 \in H^*$ such that $f^*(\chi_1) = f^*(\chi_2)$, we assume that $\chi_1 \neq \chi_2$, then there exists $h \in H$ such that $\chi_1(h) \neq \chi_2(h)$, since f is surjective, there exists $g \in G$ such that $f(g) = h$, hence $f^*(\chi_1)(g) \neq f^*(\chi_2)(g)$, so it is absurd.

If f is injective, then injective module $\mathbb{S}^1$ allows us to extend a charcter $\chi : G \rightarrow \mathbb{S}^1$ to $\tilde{\chi} : H \rightarrow \mathbb{S}^1$ such that $\tilde{\chi} \circ f = \chi$. We need to verify the continuity of $\tilde{\chi}$: notice that f is open, so $f(G)$ is the open subgroup of H , hence we can prove that $\tilde{\chi}$ is continous on $f(G)$, then we can conclue that $\tilde{\chi}$ is continous on H by translation (a group homomorphism is continous on some open subgroup, then the morphism is just continous). $\square$

Theorem 2.2 (Pontryagin-Van Kampen, 1934).

Let $G \in \mathbf{LCA}$ and G^* be its dual group, then evaluation map gives a natural ismoprhmism in $\mathbf{LCA}$ by fixing one variable:

$$\text{ev} : G \rightarrow (G^*)^*, \quad g \mapsto \text{ev}_g$$

where $\text{ev}_g : G^* \rightarrow \mathbb{S}^1$ is defined by $\text{ev}_g(\chi) = \chi(g)$.

Proof. $\square$

Here is the slogan of the duality:

every LCA-group is the dual group of its dual group.

The conclusion is similar with the finite-dimension vector space, hence it will be a strong tool to study the structure of LCA-groups and the analysis on it. In the sense of category theory, the Pontryagin duality is a contravariant equivalence with inverse itself:

$$(-)^* : \mathbf{LCA} \rightarrow (\mathbf{LCA})^{op}$$

Consequences

It is wonderful to see what we can get from the duality, we make a convention that G in this section always means a LCA-group.

Proposition 2.15. In the sense of topological isomorphism

$$G_1 \oplus G_2 \cong G_1 \times G_2$$

and

$$(G_1 \times G_2)^* \cong G_1^* \times G_2^*$$

Proof. In the sense of category of abelian groups, we have

$$(G_1 \times G_2)^* \cong G_1^* \oplus G_2^* \cong G_1^* \times G_2^*$$

So we just need to verify the isomorphism is homemomorphism under the compact-open topology. $\square$

Proposition 2.16. If G is compact, then G^* is discrete; If G is discrete, then G^* is compact.

Proof. $\square$

Proposition 2.17. If G is compact group, then

- G is metrizable if and only if G^* is countable.
- G is connected if and only if G^* is torsion-free.

3 Linear Lie groups

Matrix groups are the most important examples of Lie groups and Lie algebras, this section is some preparation for the general case.

3.1 Exponential map

Remember that $M_n(K) \cong K^{n^2}$ for any field (even any ring), in particular we consider $K = \mathbb{C}$ or $\mathbb{R}$, then the space of endomorphisms or matrices is just a finite-dimensional vector space over the field, so we can define the typical norm on it:

$$\begin{aligned}\|A\| &= \sqrt{\sum_{i,j} |a_{ij}|^2} \quad (\text{euclidean norm}) \\ \|A\|_\infty &= \max_{i,j} |a_{ij}| \quad (\text{Max norm}) \\ \|A\|_1 &= \sum_{i,j} |a_{ij}| \quad (\text{Sum norm})\end{aligned}$$

Moreover, we can consider the norm of $M_n(K)$ from another view, we view it as the linear continuous operator inside K^n , hence in particular the norm of K^n can induce the norm of operator:

$$\|A\|_{op} = \sup_{x \in K^n - \{0\}} \frac{\|Ax\|}{\|x\|}$$

Anyway, the space of endomorphisms on finite-dimensional vector space or the matrix space is still finite-dimensional, as a Banach space, all the norms on $M_n(K)$ is equivalent, with the basic knowledge of topology on it, we can talk about some whether some series in analysis makes sense or not, in particular, exponential map:

Definition 3.1. For a matrix $A \in M_n(K)$, the exponential of A is defined by

$$\exp(A) = \sum_{k \geq 0} \frac{A^k}{k!}$$

or sometimes we denote it by e^A .

The definition is analogous to the exponential map in $\mathbb{C}$, and we can verify that it is well-defined for any A by

$$\|\exp(A)\| \leq \sum_{k \geq 0} \frac{\|A\|^k}{k!} = \exp(\|A\|) < +\infty$$

so exponential map still makes sense for linear algebra! Some algebraic properties can be conclude to handle different calculation, and they are important.

Proposition 3.1.

Let $K = \mathbb{C}$ or $\mathbb{R}$, then

- (1) $ge^Xg^{-1} = e^{gXg^{-1}}$ for any $g \in \text{GL}_n(K)$ and $X \in M_n(K)$.
- (2) If $XY = YX$, then $e^{X+Y} = e^Xe^Y$.
- (3) e^X is invertible with inverse e^{-X} .

Proof. (1) It is easy to check that

$$(gXg^{-1})^k = gX^k g^{-1}$$

thus by definition we have the equality. For (2), we can expand e^{X+Y} by definition and binomial formula, and the commutativity of X and Y allows us to rearrange the terms. For (3), we see the X as the complex matrix and then by Jordan form, we have

$$gXg^{-1} = D(I + N)$$

under the base change of g , where D is diagonal and N is nilpotent such that $DN = ND$, thus

$$ge^X g^{-1} = e^D e^{I+D}$$

with e^D is diagonal and e^{I+D} an upper-triangular matrix with non-zero diagonal entries, so we can conclude that e^X is invertible with inverse e^{-X} by the property (2). $\square$

There are many other proof of the proposition, and each of them gives a new view about the matrix group and Lie group, for example, we can consider the exponential map as the solution of ODE:

Theorem 3.1. For any $X \in M_n(K)$, the one-parameter

$$\Phi_X : \mathbb{R} \rightarrow M_n(K), \quad t \mapsto e^{tX}$$

is the unique **global** C^1 **solution** of the initial value problem of ODE:

$$\frac{d}{dt}Y(t) = X\Phi_Y(t), \quad \Phi_X(0) = I$$

with Y the matrix-valued function of t .

Proof. The uniqueness of the solution can be proved by Cauchy-Lipschitz theorem, for the existence, we should verify that Φ_X is exactly the solution: For any $t \in \mathbb{R}$

$$\frac{\Phi_X(t+h) - \Phi_X(t)}{h} = \frac{e^{(t+h)X} - e^{tX}}{h} = e^{tX} \frac{(e^{hX} - I)}{h}$$

The convergence of the exponential series is uniform, which allows us to exchange the limit and the sum, thus we can deduce the derivative of Φ_X is exactly $X\Phi_X$, hence Φ_X is the solution of the ODE (surely it is C^1 , Φ_X is derivative so it is continuous, that implies that $X\Phi_X$ is continuous too). $\square$

Remark 3.1. For property (2) in proposition 3.1, here we can consider the ODE

$$Z'(t) = (X + Y)Z(t), \quad Z(0) = I$$

we can verify that $t \mapsto e^{X+Y}$ and $t \mapsto e^X e^Y$ are both solutions when X and Y commute, thus they are equal by the uniqueness of the solution.

Remark 3.2. Actually, the theorem contain an simple information is that

$$\frac{d}{dt}(e^{tX}) = Xe^{tX}$$

so we notice the amazing similarity with the case in euclidean space, which allows us to consider the one-parameter subgroup as a dynamic system, and the exponential map is the flow of the system, this is the original ideal of the exponential map in Lie theory.

Another method to prove that e^X is invertible is to consider its determinant, which is natural to motivate the following result:

Theorem 3.2 (Jacobi's formula). For any $A \in M_n(K)$, we have

$$\det(e^A) = e^{\text{tr}(A)}$$

Proof. By Schur's lemma, any $X \in M_n(\mathbb{C})$ can be upper-triangularized by some $g \in GL_n(\mathbb{C})$ such that $X = gTg^{-1}$ the upper-triangular matrix T with diagonal entries $\lambda_1, \dots, \lambda_n$, thus

$$\det(e^X) = e^{\lambda_1} \dots e^{\lambda_n} = e^{\lambda_1 + \dots + \lambda_n} = e^{\text{tr}(X)}$$

□

Another analytic aspects of the exponential map is asymptotic expansion, that means we can manipulate the exponential map as what we do in analysis, but first of all we prove that the exponential map is similarly a smooth map by the requirement of the expansion:

Proposition 3.2. For any $X \in M_n(K)$, the one-parameter

$$\Phi_X : \mathbb{R} \rightarrow GL_n(K), \quad t \mapsto e^{tX}$$

defines a real-analytic homomorphism, and we have the following expansion

$$(1) \exp(tX) \exp(tY) = \exp(t(X + Y) + \frac{1}{2}[X, Y] + O(t^3))$$

$$(2) \exp(tX) \exp(tY) \exp(-tX) \exp(-tY) = \exp(t^2[X, Y] + O(t^3))$$

for any $X, Y \in M_n(K)$, with bracket $[X, Y] = XY - YX$ the commutator of X and Y .

Proof. For any $t_0 \in \mathbb{R}$, we can expand Φ_X at t_0 by definition:

$$e^{tX} = e^{t_0X} e^{(t-t_0)X} = e^{t_0X} \sum_{k=0}^{\infty} \frac{(t-t_0)^k}{k!} X^k$$

The series converges uniformly on any compact interval, so the map is real-analytic. To prove the formula, we just need to expand the left-hand side by definition and rearrange the terms. □

Remark 3.3. If $XY = YX$, then the commutator $[X, Y] = 0$, thus the expansion in (1) is exact, which is just the property (2) in proposition 3.1, hence we get the same result by using the expansion.

Above properties of one-parameter allows us to consider the general case in matrix group, later we will see that the matrix group is indeed a smooth manifold, i.e. a Lie group.

Theorem 3.3. Let $\gamma : \mathbb{R} \rightarrow GL_n(\mathbb{R})$ be a one-parameter subgroup, then there exists a unique $X \in M_n(\mathbb{R})$ such that $\gamma(t) = e^{tX}$ for any $t \in \mathbb{R}$, so any one-parameter subgroup of $GL_n(\mathbb{R})$ is exactly real-analytic.

Proof. convolution... □

3.2 General linear group

Formally, we define the real-general linear group as following:

$$GL_n(\mathbb{R}) := \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}$$

it is the subset of $M_n(\mathbb{R})$, while the topology of $GL_n(\mathbb{R})$ is just the subspace topology induced by $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$, hence we can conclude something about the topology of the group:

- $GL_n(\mathbb{R})$ is a open subset of $M_n(\mathbb{R})$, it is immediately by

$$GL_n(\mathbb{R}) = \det^{-1}(\mathbb{R} - \{0\})$$

since determinant is a continuous map, so it can be viewed as a open subset of euclidean space $\mathbb{R}^{n^2}$, so it is a smooth manifold.

- Following the above result, since it can be viewed as an open subset of euclidean space, so it is a "bon espace" in the sense of topology, i.e. semi-locally simply connected, locally path-connected, locally compact, and Hausdorff.
- Consider the operation of the group

$$m(A, B) = AB^{-1} = \frac{1}{\det(B)} A \cdot \text{adj}(B)$$

where $\text{adj}(B)$ is the adjoint matrix of B , which is a polynomial of entries of B , hence this map is smooth as polynomial map, which shows that $GL_n(\mathbb{R})$ is a topological group, and in particular, with the good smooth structure, it is a Lie group.

- As the topological group, the local structure is same, so we can just consider the point $I \in GL_n(\mathbb{R})$. As a differential manifold, we can consider the tangent space at I , so we can identify the tangent space

$$T_I(GL_n(\mathbb{R})) \cong \mathbb{R}^{n^2} \cong M_n(\mathbb{R})$$

Actually, we can view a strong result: For any $X \in M_n(\mathbb{R})$, we define a curve on manifold by

$$\gamma_X : \mathbb{R} \rightarrow GL_n(\mathbb{R}), \quad t \mapsto e^{tX}$$

by exponential map, this is smooth curve with derivative $\gamma'_X(0) = X$, hence we prove that $M_n(\mathbb{R})$ is exactly the tangent space of $GL_n(\mathbb{R})$ at I , and we realize the connection between the local structure of the Lie group and its tangent space by one-parameter subgroup, that is the basic ideal of Lie theory.

We conclude above discussion by the following definition, or exactly the **vocabulary**, because they are just the new notation of manifold and vector space, but they are the basic object of Lie theory, so we need to give them a name:

Definition 3.2. A **lie group** is a smooth manifold G with a group structure such that the group operation $m : G \times G \rightarrow G$ defined by $m(g, h) = gh^{-1}$ is smooth.

A **lie algebra** is a $\mathbb{R}$ -vector space $\mathfrak{g}$ with a bilinear map $[-, -] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ such that for any $x, y, z \in \mathfrak{g}$ and $a, b \in \mathbb{R}$, we have

$$(1) [x, y] = -[y, x] \text{ (antisymmetry)}$$

$$(2) [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0 \text{ (Jacobi identity)}$$

with above discuss we can see that $GL_n(\mathbb{R})$ is exactly a Lie group, while its tangent space at identity $M_n(\mathbb{R})$ is a vector space, actually it is a Lie algebra by defining the bracket as following:

$$[X, Y] = XY - YX$$

and we say $\mathfrak{h}$ is a **sub(-lie) algebra** of $\mathfrak{g}$ if $\mathfrak{h}$ is a subspace of $\mathfrak{g}$ and is closed under the bracket, i.e.

$$\forall x, y \in \mathfrak{h}, [x, y] \in \mathfrak{h}$$

To study other matrix groups, we can just consider the submanifold of $GL_n(\mathbb{R})$, i.e. the subgroup of $GL_n(\mathbb{R})$ with the subspace topology, hence we can get the following result:

Theorem 3.3. Let G be a closed subgroup of $GL_n(\mathbb{R})$, then G is a submanifold of $GL_n(\mathbb{R})$, and the tangent space of G at identity is a Lie algebra with

$$T_I(G) = \{M \in M_n(\mathbb{R}) \mid e^{tM} \in G, \forall t \in \mathbb{R}\}$$

it is a **subalgebra** of $M_n(\mathbb{R})$ with the bracket defined by commutator.

Proof. □

which motivates the following definition:

Definition 3.1. A **linear lie group** is a closed subgroup of $GL_n(\mathbb{R})$ for some n , and its lie algebra is defined as

$$\text{Lie}(G) = \{M \in M_n(\mathbb{R}) \mid e^{tM} \in G, \forall t \in \mathbb{R}\}$$

and we usually denote it by frak character $\mathfrak{g}$.

We invite the Lie algebra of $GL_n(\mathbb{R})$ by consider its tangent space at identity, conversely, exponential map allows us to know the local structure of the group by its Lie algebra, which can be done beyond the general manifold

Theorem 3.4. The exponential map $\exp : M_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ is a local diffeomorphism at 0, in particular, it is a diffeomorphism from some neighborhood of 0 to the image.

Proof. □

Let us consider some examples related to the general linear group:

Example 3.1. We view $\mathbb{C}$ as a real vector space, then there exists a natural embedding

$$i : M_n(\mathbb{C}) \rightarrow M_{2n}(\mathbb{R}), \quad A + iB \mapsto \begin{pmatrix} A & -B \\ B & A \end{pmatrix}$$

it is a injective continous linear map, and we can verify that

$$(A + iB)(x + iy) = 0 \iff \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

which shows that restriction of i on $GL_n(\mathbb{C})$ takes image in $GL_{2n}(\mathbb{R})$, and image $i(GL_n(\mathbb{C}))$ is indeed a closed subgroup of $GL_{2n}(\mathbb{R})$, hence it is a linear lie group, and its lie algebra is exactly the image of i on $M_n(\mathbb{C})$, which motivates the following diagramme

$$\begin{array}{ccc} M_n(\mathbb{C}) & \xrightarrow{\iota} & M_{2n}(\mathbb{R}) \\ \exp \downarrow & & \downarrow \exp \\ GL_n(\mathbb{C}) & \xrightarrow{\iota} & GL_{2n}(\mathbb{R}) \end{array}$$

similarly with general linear group, the closed subgroup of $GL_n(\mathbb{C})$ is also a submanifold of $GL_n(\mathbb{C})$, and the tangent space at identity is the subalgebra of $M_n(\mathbb{C})$ with the bracket defined by commutator, hence we can get a large class of linear lie groups and lie algebras by this method, it is a big case of linear lie groups.

Example 3.2. (Affine group) We consider a closed subgroup of $GL_{n+1}(\mathbb{R})$ defined by

$$G = \left\{ \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \mid A \in GL_n(\mathbb{R}), b \in \mathbb{R}^n \right\}$$

and we consider the group action $G \curvearrowright \mathbb{R}^n$ defined by

$$\begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \cdot x = \begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} = Ax + b$$

This action is transitive so $\mathbb{R}^n$ is a homogeneous space under G , and we can establish a homeomorphism

$$G/GL_n(\mathbb{R}) \cong \mathbb{R}^n$$

It is a classic example about Erlangen programme, here G is a important geometric group, we usually call it the **affine group** $\text{Aff}_n(\mathbb{R})$.

3.3 Adjoint map

Review a bsic fact about the finite-dimensional vector space: any norm on a finite-dimensional Banach space is equivalent, hence it is natural to consider the abstract exponential map on linear opreators

Definition 3.3. Let V be a finite-dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$, we define the exponential map $\text{Exp} : \text{End}(V) \rightarrow GL(V)$ by

$$\text{Exp}(T) = \sum_{k \geq 0} \frac{T^k}{k!}$$

where $T^k = T \circ \dots \circ T$ is the k -th composition of T .

The definition is well-defined when we consider $\text{End}(V)$ as a Banach space, and we can verify that the properties in proposition 3.1 still hold for the abstract exponential map. In particular, if we choose a basis of V and denote it by $\mathcal{B}$, then we can see their connection by the following commutative diagram:

$$\begin{array}{ccc} \text{End}(V) & \xrightarrow{\text{Exp}} & \text{GL}(V) \\ \downarrow [\cdot]_{\mathcal{B}} & & \downarrow [\cdot]_{\mathcal{B}} \\ M_n(K) & \xrightarrow{\text{exp}} & \text{GL}_n(K) \end{array}$$

or simply the equality

$$[\text{Exp}(T)]_{\mathcal{B}} = \text{exp}([T]_{\mathcal{B}})$$

We consider the conjugation action of a linear lie group G on itself, and then for any $g \in G$, there exists a diffeomorphism by the smooth structure of G

$$c_g : G \rightarrow G, \quad h \mapsto ghg^{-1}$$

with $c_g(I) = I$, hence it allows us to consider the differential of c_g at identity,

$$(Dc_g)_I : T_I(G) \rightarrow T_I(G), \quad X \mapsto gXg^{-1}$$

the reason is as following: for linear lie group, the tangent space at identity is just $\text{Lie}(G)$ by theorem 3.3, so for any $X \in \text{Lie}(G)$, there exists a one-parameter subgroup $\gamma_X : \mathbb{R} \rightarrow G$ defined by $\gamma_X(t) = e^{tX}$ with $X = \frac{d}{dt}\big|_{t=0} \gamma_X(t)$, hence we consider the parameterization of the conjugation

$$\frac{d}{dt}\big|_{t=0} c_g(\gamma_X(t)) = \frac{d}{dt}\big|_{t=0} g\gamma_X(t)g^{-1} = gXg^{-1}$$

which shows that $(Dc_g)_I(X) = gXg^{-1}$ exactly, hence by the construction we induce a representation

$$\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g}), \quad g \mapsto \text{Ad}_g = (Dc_g)_I$$

and we call the representation Ad the **adjoint representation** of G .

Notice that $\mathfrak{g}$ is finite-dimensional, so $\text{GL}(\mathfrak{g})$ is also a linear Lie group, with its Lie algebra $\text{End}(\mathfrak{g})$, hence we can consider the differential of Ad at identity

Proposition 3.5. Let G be a linear lie group with lie algebra $\mathfrak{g}$, then the direction derivative of the adjoint representation at identity is

$$(D \text{Ad})_I(X) = \frac{d}{dt}\big|_{t=0} \text{Ad}_{\exp(tX)} = \text{ad}_X$$

with $\text{ad}(X) : \mathfrak{g} \rightarrow \mathfrak{g}$ defined by $\text{ad}_X(Y) = XY - YX$ the commutator of X and Y .

Proof. caculation... □

Hence we get the basic idea of the adjoint

Definition 3.4. Let G be a linear lie group with its lie algebra $\mathfrak{g}$, then

- The adjoint representation of G is defined by

$$\text{Ad} : G \rightarrow \text{Aut}(\mathfrak{g}), \quad g \mapsto \text{Ad}_g$$

with $\text{Ad}_g(X) = gXg^{-1}$ for any $X \in \mathfrak{g}$.

- The adjoint representation of Lie algebra $\mathfrak{g}$ is defined by

$$\text{ad} : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g}), \quad X \mapsto \text{ad}_X$$

with $\text{ad}_X(Y) = [X, Y]$ for any $Y \in \mathfrak{g}$.

Remark 3.4. An application between two lie algebras $f : \mathfrak{g} \rightarrow \mathfrak{h}$ is a **morphism of Lie algebra** if it is a linear map and preserves the bracket

$$f([X, Y]_{\mathfrak{g}}) = [f(X), f(Y)]_{\mathfrak{h}}$$

In particular, the isomorphism in the category of lie algebras is called **automorphism**, and we use $\text{Aut}(\mathfrak{g})$ to denote the group of automorphisms of $\mathfrak{g}$, which is subspace of $\text{GL}(\mathfrak{g})$, the latest one is the set of all bijective linear maps on $\mathfrak{g}$.

with above discussion, we can describe a core properties of the adjoint representation, which is the connection between the local structure of the group and its lie algebra:

Theorem 3.4. Let G be a linear lie group with its lie algebra $\mathfrak{g}$, then for any $X \in \mathfrak{g}$, we have

$$\text{Ad}_{\exp(X)} = \text{Exp}(\text{ad}_X)$$

i.e. the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\text{ad}} & \text{End}(\mathfrak{g}) \\ \exp \downarrow & & \downarrow \text{Exp} \\ G & \xrightarrow{\text{Ad}} & \text{GL}(\mathfrak{g}) \end{array}$$

Proof.

□

4 Field and Galois Theory

To study the structure of the field, consider a natural map: Let K be a field

$$\phi : \mathbb{Z} \rightarrow K, \quad 1 \mapsto 1_K$$

It is a well-defined homomorphism, which invites a ideal of $\mathbb{Z}$

$$\ker \phi = \{n \in \mathbb{Z} | n \cdot 1_K = 0_K\}$$

By the structure of $\mathbb{Z}$, so there exists a integer $p \in \mathbb{Z}$ such that $\ker \phi = p\mathbb{Z}$, which gives the definition of the characteristic of the field.

Definition 4.1. For any field K , we define $\text{char}(K)$ be the characteristic of the field: either 0 or the smallest integer $n \in \mathbb{N}$ such that $n \cdot 1_K = 0$, by the natural map equivalently

$$\begin{cases} \text{char}(K) = 0 \iff \text{im} \phi \cong \mathbb{Z} \\ \text{char}(K) = p \iff \text{im} \phi \cong \mathbb{Z}/p\mathbb{Z} \end{cases}$$

The characteristic of a field is either zero or a prime number. Suppose that $\text{char}(K) = p \neq 0$, if $p = ab$ is not a prime, then we will get two zero divisor $a \cdot 1_K$ and $b \cdot 1_K$, but a field can not contain any zero divisor, so p must be prime. An amazing thing in a field (or ring) with non-zero characteristic p is that we can write a equation

$$(x + y)^p = x^p + y^p$$

it has an interesting name: **freshman's dream**, this is an equality that would be written by someone who has studied very little mathematics or is just beginning to learn it.

Another natural map is **Frobenius endomorphism**, let K be a field with non-zero characteristic p , then we define

$$\sigma : K \rightarrow K, \quad x \mapsto x^p$$

It is a well-defined injective field homomorphism, so it is a endomorphism, but it is not necessary surjective.

Example 4.1. Consider $K = \mathbb{F}_p(x)$ a field of rational functions. $\text{char}(K) = p$ since $\text{im} \phi = \mathbb{F}_p$, and it is easy to observe that there exists no $f \in \mathbb{F}_p(x)$ such that $f^p(x) = x$, so the Frobenius map here is not surjective.

A good case is finite field, in that case Frobenius endomorphism is furthermore a automorphism, so the Frobenius map will be in Galois group and it reflects the structure of the finite field.

4.1 Finite extension

For the beginning, notice a classic isomorphism:

$$\mathbb{R}[X]/(X - a) \cong \mathbb{R}$$

For any $a \in \mathbb{R}$, and we review the isomorphism given in complex number field:

$$\mathbb{R}[X]/(X^2 + 1) \cong \mathbb{C}$$

Clearly, $\mathbb{C}$ is a larger field containing $\mathbb{R}$, but we wonder how we can get it from the algebra structure. Here it motivates us to consider the question from the polynomial structure defined on a field. Firstly we need some lemmas, it is very clear after commutative ring:

Lemma 4.1. Let L be a field containing K as a subfield, then L is a vector space over K .

This lemma allows us to give a rough classification of field extension:

Definition 4.2. If L is a field containing K as a subfield.

- L/K or $K \subset L$ is denoted a **field extension** of K .
- L/K is a **finite extension** if L is a finite-dimensional vector space over K . The **degree** of the finite extension is defined as $[L : K] := \dim_K L$.
- $a \in L$ is **algebraic** over K if there exists a non-zero polynomial $f \in K[X]$ such that $f(a) = 0$, L/K is an **algebraic extension** if every element in L is algebraic over K .
- a is **transcendental** over K if it is not algebraic over K .

Another lemma is the key to construction of field, the reason is that in a PID ring a ideal is maximal if and only if it is generated by an irreducible element.

Lemma 4.2. If K a field and $p \in K[X]$, then p is irreducible if and only if $K[X]/(p)$ is a field.

Hence the a contrsuction of field extension can be given by the quotient of polynoimal ring over a field by a good choice of ideal:

Proposition 4.3. Let K be a field and I be a principiapl ideal generated by a monic irreducible polynoimal $p \in K[X]$ of degree d , Let $L = K[X]/I$, then

- (1) L is a field and K can be embedded in L , so K can be identified as a subfield of L .
- (2) p has a root β in L , exactly $\beta = X + I \in L$
- (3) If $g \in K[X]$ and β is a root of g in L , then $p|g$.
- (4) p is the unique monic irreducible polynoimal in $K[X]$ having β as a root in L .
- (5) L is a finite extension of K of degree d with the basis $\{1, \beta, \dots, \beta^{d-1}\}$.

Proof. (1) L is a field by above lemma, and we consider an embedding $i(a) = a + I$ for any $a \in K$, clearly it is injective and actually it is $\pi|_K$, where π is the canonical map from $K[x]$ to $K[x]/I$.

(2) Suppose $p(x) = a_0 + \dots + a_d x^d$, then in L , we have

$$\begin{aligned} p(\beta) &= a_0 + a_1 \beta + \dots + a_d \beta^d \\ &= a_0 + a_1 (X + I) + \dots + a_d (X + I)^d \\ &= a_0 + a_1 (X + I) + \dots + a_d (X^d + I) \\ &= p(X) + I = I \end{aligned}$$

Here $p(X) \in I$ and I is the zero element in L .

(3) If p does not divide g , then $\gcd(p, g) = 1$ since p is irreducible, so PID ring $K[X]$ implies the existence of r, t in $K[X]$ such that

$$1 = s(x)p(x) + t(x)g(x)$$

put $x = \beta$ in L , then $1 = 0$ leads to a contradiction.

(4) immediately from (3). For (5) we use euclidean division for polynomial, any $f \in K[X]$, there exists $q, r \in K[X]$ with $\deg r < d$ such that $f(x) = q(x)p(x) + r(x)$, then $f + I = r + I$ in L , and if $r(x) = b_0 + \dots + b_k x^k$, $k < d$, by operations of ideal

$$r + I = b_0 + b_1 \beta + \dots + b_k \beta^k$$

so $\{1, \beta, \dots, \beta^{d-1}\}$ spans L . They are linearly independent because if we assume they are linearly dependent, that means there exists a polynomial $h \in K[X]$ of degree $< d$ such that h has β as the root, but by (3) we know that $p|h$, which leads to a contradiction since $d = \deg p \leq \deg h$. $\square$

Review the example in the beginning, the complex number field can also be constructed by adjoining element i to $\mathbb{R}$ such that $i^2 + 1 = 0$, and we always write a complex number of the form $a + ib$ with a, b real numbers, which is another construction of field extension, and more generally we have the following proposition:

Proposition 4.4. Let K be a field and $f \in K[X]$ irreducible, adjoin element $c \notin K$ such that c is a root of f , then $K[c]$ is a field containing K as a subfield, and it can be written as $K(c)$.

$$K(c) = \{a_0 + a_1 c + \dots + a_k c^k \mid a_1, \dots, a_k \in K, k = \deg f - 1\}$$

Proof. $K[c]$ naturally is a ring with same identity with K , so we just need to find the inverse. For any polynomial $p \in K[X]$ with degree less than $\deg f$, it is co-prime with f since f is irreducible, so by Bezout's theorem for polynomial, there exists $r, t \in K[X]$ such that

$$p(x)r(x) + t(x)f(x) = 1$$

hence we take $x = c$ then immediately $p(c)r(c) = 1$, so in $K[c]$ we find the inverse of $p(c)$. and notice that $K[c]$ is a field guarantee any rational polynomial can have the form of polynomial, so $K[c] = K(c)$ evidently. $\square$

Here we find two method to extension the field, the first method is consider the quotient of polynomial ring, and we can embed K into it; the second is more direct, we just add some element in the field.

Theorem 4.1 (Structure of field extension).

Let L/K be field extension of K and $a \in L$ is algebraic, then

- (1) There exists a unique **monic** irreducible polynomial in $K[X]$ having a as a root, formally we call it **minimal polynomial** of a over K , and denote it by Π_a .
- (2) If $I = (\Pi_a)$, then there exists an isomorphism

$$\phi : K[X]/I \rightarrow K(a)$$

with $X + I \mapsto a$ and $c + I \mapsto c$ for all $c \in K$.

- (3) Let L'/K is another field extension and $a' \in L'$ is also a root of Π_a , then there exists an isomorphism $\psi : K(a) \rightarrow K(a')$ such that $\psi|_K = \text{id}_K$ and $\psi(a) = a'$.

Proof. It needs to consider the **valuation map**, we define $h : K[X] \rightarrow L$ by $p \mapsto p(a)$, then clearly $\ker h$ contains all polynomials having a as a root, then notice that $K[X]$ is a principal ideal ring, so $\ker h$ must be the form (p) with monic $p \in K[X]$. By first

isomorphism theorem, we know that $K[X]/I \cong K(a)$, then by Lemma 4.2 p must be irreducible, hence we prove (1) and (2), and we can draw commute diagram to finish (3):

$$\begin{array}{ccc}
 K[X] & \xrightarrow{h} & K[a] \\
 \downarrow h' & \searrow \pi & \uparrow \simeq \\
 K[a'] & \xleftarrow{\simeq} & K[X]/I
 \end{array}$$

where $h'(p) = p(a')$, and π is the canonical map. □

Remark 4.1. The theorem shows a induction

$$\{\text{algebraic elements in } L/K\} \rightsquigarrow \{\text{monic irreducible polynomial over } K\}$$

conversely it is not right since the field extension L may be not enough large, which refers the "**closure**" in the sense of algebraic element. For statement (3), a simple example is $\mathbb{Q}(\sqrt{2}) = \mathbb{Q}(-\sqrt{2})$, and monic irreducible polynomial $X^2 - 2 \in \mathbb{Q}[X]$.

Trace, norm and Discriminant

For any finite extension L/K of degree n , we can view element $a \in L$ as a K -linear map

$$m_a : L \rightarrow L, \quad x \mapsto ax$$

in particular, it is an automorphism of L if $a \neq 0$, and it satisfies several properties:

- $m_{a+b} = m_a + m_b$
- $m_{ab} = m_a \circ m_b$
- $m_{c \cdot a} = c \cdot m_a$

with $a, b \in L$ and $c \in K$. Since L is a finite-dimensional, it is natural to consider m_a as a matrix and define two invariants in linear algebra

- Trace of a over K : $\text{Tr}_{L/K}(a) = \text{Tr}(m_a)$
- Norm of a over K : $N_{L/K}(a) = \det(m_a)$

exactly what we do can be concluded as the following

$$L \hookrightarrow \text{End}_K(L) \rightarrow K$$

the first map refers to a **regular representation** of L over K , and the second map is what we are familiar with, and we can verify some properties of trace and norm:

- $\text{Tr}_{L/K}(a + b) = \text{Tr}_{L/K}(a) + \text{Tr}_{L/K}(b)$
- $\text{Tr}_{L/K}(c \cdot a) = c \cdot \text{Tr}_{L/K}(a)$
- $N_{L/K}(ab) = N_{L/K}(a) \cdot N_{L/K}(b)$

- $N_{L/K}(c \cdot a) = c^n \cdot N_{L/K}(a)$
- $\text{Tr}_{L/K}(c) = nc$ and $N_{L/K}(c) = c^n$

with $a, b \in L$ and $c \in K$. We can also give a more explicit description of trace and norm by the minimal polynomial of a over K , moreover, for any endomorphism m_a we can consider its **characteristic polynomial** $\chi_a(t) = \det(t \cdot \text{id}_L - m_a)$, and conclude the following result:

Proposition 4.5. For any L/K finite extension with degree n and $a \in L$,

(a) $\chi_a = (\Pi_a)^d$ with $d = [L : K(a)]$.

(b) Suppose $\Pi_a(t) = t^m + c_{m-1}t^{m-1} + \dots + c_0$, then

$$\text{Tr}_{L/K}(a) = -dc_{m-1} \quad N_{L/K}(a) = (-1)^n c_0^d$$

Proof. (a) It is clear that $\deg \Pi_a = [K(a) : K]$, we denote it by m , then

$$\Pi_a(t) = t^m + c_{m-1}t^{m-1} + \dots + c_0$$

for some $c_i \in K$, and $\{1, a, \dots, a^{m-1}\}$ is a basis for $K(a)/K$, we suppose that $\{e_1, \dots, e_d\}$ is a basis for $L/K(a)$, then we can conclude a basis of L/K is

$$e_1, \dots, e_1 a^{m-1} \mid e_2, \dots, e_2 a^{m-1} \mid \dots \mid e_d, \dots, e_d a^{m-1}$$

so we can write the matrix of m_a under this basis

$$m_a \sim \text{diag}(A_1, \dots, A_d), \quad A_i = \begin{pmatrix} 0 & 0 & \dots & -c_0 \\ 1 & 0 & \dots & -c_1 \\ 0 & 1 & \dots & -c_2 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -c_{m-1} \end{pmatrix}$$

By calculate the characteristic polynomial of A_i we can get exactly

$$\chi_a(t) = \prod_{i=1}^d \det(t \cdot I - A_i) = (\Pi_a(t))^d$$

(b) By vieta's formula we can conclude that

$$\chi_a(t) = t^n - \text{Tr}_{L/K}(a)t^{n-1} + \dots + (-1)^n N_{L/K}(a)$$

then by formula of (a) we can calculate the result. □

4.2 Algebraic Closed field

4.3 Normal extension and Automorphism

Example 4.1. We consider the field extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$, here the minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is $X^3 - 2$, we can solve it in $\mathbb{C}$ with roots:

$$\sqrt[3]{2}, \quad \sqrt[3]{2}\omega, \quad \sqrt[3]{2}\omega^2 \quad \text{with } \omega = e^{2\pi i/3}$$

by above theorem we can conclude $\mathbb{Q}(\sqrt[3]{2}) \cong \mathbb{Q}(\sqrt[3]{2}\omega) \cong \mathbb{Q}(\sqrt[3]{2}\omega^2)$, and it also implies a **question**: quotient ring such as $\mathbb{Q}[X]/(X^3 - 2)$ can not construct a field containing all roots of irreducible polynomial such as $X^3 - 2$.

However, we can construct a larger field since $\mathbb{Q}(\sqrt[3]{2})$ has been a field: equivalently, we can adjoin ω to $\mathbb{Q}(\sqrt[3]{2})$, or consider the irreducible polynomial $X^2 + X + 1$ over $\mathbb{Q}(\sqrt[3]{2})$, then we can get a larger field $\mathbb{Q}(\sqrt[3]{2}, \omega)$ containing all roots of $X^3 - 2$.

Above is a typical example to motivate the definition of splitting field, and we can give a more general definition:

Suppose L/K is a extension and a polynomial $f \in K[X]$, we say f **splits** over K if there exists $c, a_1, \dots, a_n \in K$ such that f can be factorized as linear terms:

$$f(x) = c(x - a_1)(x - a_2) \cdots (x - a_n)$$

above examples shows that a extension L/K can not always contain all roots of a polynomial, and we can construct a larger field.

Theorem 4.6 (splitting field).

Let K be a field and $f \in K[X]$ with degree $n \geq 1$, then there exists a smallest field extension L/K such that f splits over L , formally such a field extension is called **splitting field** of f over K , and it satisfies:

- $L \cong K[r_1, \dots, r_n]$ with $r_1, \dots, r_n$ are the roots of f in L , it shows that the splitting field is unique up to isomorphism.

- As a finite extension, $[L : K] \leq n!$

Proof. □

Remark 4.2. L.Kronecker proved that for any field K and $f \in K[X]$, there exists a extension L/K such that f splits over L . The smallest of such extension is the splitting field of f , and an extremely larger field is the algebraic closed field of K , although the result is a little old in nowadays.

For the example at the begining, the experience of $\mathbb{C}$ gives us a hint that $\mathbb{Q}(\sqrt[3]{2})$ can not contain the complex number ω , which reminds us to consider the roots of the polynomial in algebraic closure.

Definition 4.1. Let L/K be an algebraic extension, we define

$$\text{Hom}_{K\text{-alg}}(L, \bar{K}) = \{f : L \hookrightarrow \bar{K} \mid f|_K = \text{id}_K\}$$

to be **K -embedding of L** .

Remark 4.3. It is natural to consider the category of field extensions of K , we denote it by $\mathbf{Ext}_K$. Its objects are the extension $K \subset L$, or we can see it as a K -algebra by an embedding $\tau : K \hookrightarrow L$. Hence its morphism is just the morphism of K -algebra, for example if we consider $f : L \rightarrow L'$, then f must be injective since it is a morphism of field, and $f(r) = rf(1) = r$ for any $r \in K$, in particular $L' = \bar{K}$ we get the definition of K -embedding of L . Hence here we get a (non-full) subcategory of $\mathbf{CAlg}_K$.

For any $L/\mathbb{Q}$ algebraic extension, we consider f is a embedding, then we have

$$f(X^3 - 2) = f(X)^3 - 2 \quad f(X^2 + X + 1) = f(X)^2 + f(X) + 1$$

for any $X \in L$. which allows us to calculate that

$$\mathrm{Hom}_{\mathbb{Q}\text{-alg}}(\mathbb{Q}(\sqrt[3]{2}), \bar{\mathbb{Q}}) = \{f_1, f_2, f_3\}$$

with $f_i(\sqrt[3]{2}) = \sqrt[3]{2}\omega^{i-1}$ for $i = 1, 2, 3$.

$$\mathrm{Hom}_{\mathbb{Q}\text{-alg}}(\mathbb{Q}(\sqrt[3]{2}, \omega), \bar{\mathbb{Q}}) = \{f_{1,1}, f_{1,2}, f_{1,3}, f_{2,1}, f_{2,2}, f_{2,3}\}$$

with $f_{i,j}(\sqrt[3]{2}) = f_i(\sqrt[3]{2})$ and $f_{i,j}(\omega) = \omega^j$ for $i = 1, 2, 3$ and $j = 1, 2$. We can observe that the second case forms exactly a group since the image of $f_{i,j}$ is $\mathbb{Q}(\sqrt[3]{2}, \omega)$ itself, and in particular it can be identified as S_3 (notice $f_{2,1}$ is of order 3 and $f_{1,2}$ is of order 2, they are not commute).

Definition 4.2. Let L/K be a field extension, we define

$$\mathrm{Gal}(L/K) = \mathrm{Hom}_{K\text{-alg}}(L, L) = \{\sigma \in \mathrm{Aut}(L) \mid \sigma|_K = \mathrm{id}_K\}$$

to be the **Galois group** (or **automorphism group**) of L/K , it is a subgroup of $\mathrm{Aut}(L)$ which fixes K .

Notice a clear inclusion

$$\mathrm{Gal}(L/K) \subset \mathrm{Hom}_{K\text{-alg}}(L, \bar{K})$$

which is sufficient to prove that when $L = \mathbb{Q}(\sqrt[3]{2}, \omega)$, the inclusion turns to be an equality. while $L = \mathbb{Q}(\sqrt[3]{2})$, we can only get $\mathrm{Gal}(L/\mathbb{Q}) = \{f_1\}$, hence the difference between Galois group and the embeddings is the key to distinguish the extensions.

Theorem 4.2. Let L/K be a algebraic extension, then the following conditions are equivalent:

1. L is a splitting field of a set of polynomial in $K[X]$.
2. $\mathrm{Gal}(L/K) = \mathrm{Hom}_K(L, \bar{K})$
3. For any $a \in L$, the minimal polynomial $\Pi_a \in K[X]$ splits over L .

If one of the above conditions is satisfied, we say L/K is a **normal extension**.

Remark 4.4. Let $a \in L$ and Π_a is the minimal polynomial of a , then K -embedding induces a bijection to the roots of Π_a in $\bar{K}$, denote the set by R_a

$$\mathrm{Hom}_K(L, \bar{K}) \rightarrow R_a, \quad f \mapsto f(a)$$

exactly

4.4 Separable extension

Example 4.2. Let us consider the function field $K = \mathbb{F}_p(t)$

- $X^p - t$ is irreducible in $K[X]$ since K is the fraction field of $\mathbb{F}_p[t]$, and

$$\mathbb{F}_p[t][X] \cong \mathbb{F}_p[t, X] \cong \mathbb{F}_p[X, t]$$

allow us to prove the irreducibility in $\mathbb{F}_p[t][X]$, so it holds in fraction field as well.

- consider $L = K[X]/(X^p - t)$, then we can get a extension $L \cong \mathbb{F}_p(t^{1/p})$, which allows to factorize the minimal polynomial

$$X^p - t = (X - t^{1/p})^p$$

That shows L is a splitting field of $X^p - t$ over K , so L/K is a normal extension of degree p .

- We exactly construct a extension L/K such that the minimal polynomial has repeated root, in this case the $\bar{K}$ -embedding fails to distinguish the roots and $\text{Gal}(L/K) = \text{Hom}_K(L, \bar{K}) = \{id\}$.

Review the case in $\mathbb{R}[X]$, although we do not prove the FTA exactly, but the our experience in $\mathbb{C}$ enough to conclude that any irreducible polynomial in $\mathbb{R}[X]$ can not have multiple roots in $\mathbb{C}$, since for the given distinct roots $R = \{r_1, \dots, r_k\}$ with multiplicity $\{m_1, \dots, m_k\}$, we can uniquely determine the monic polynomial of the form

$$\prod_{r_i \in \mathbb{R} \cap R} (X - r_i)^{m_i} \cdot \prod_{\text{otherwise}} (X^2 + b_j x + c_j)^{m_j}$$

since the complex roots always appear in pairs, so we can roughly conclude the result. Even more, we can not find a counterexample in $\mathbb{Q}[X]$ as well, this section is to distinguish such a field.

Definition 4.3. Let A be a k -algebra, an endomorphism $\partial : A \rightarrow A$ is called a **derivation** if it satisfies the Leibniz rule:

$$\forall f, g \in A, \quad \partial(fg) = \partial(f)g + f\partial(g)$$

It is easy to verify that

$$\begin{aligned} \partial(\lambda) &= 0, \quad \forall \lambda \in k \\ \partial(f^n) &= n f^{n-1} \partial(f), \quad \forall f \in A, n \in \mathbb{N} \end{aligned}$$

We can use the derivation to define the formal derivative of a polynomial, which is a good view that does not depend on the the exactly form of the polynomial.

$$\Omega_{L/K} = 0 \iff L/K \text{ is separable}$$

This is the general explanation of the separable extension by Kähler differentials. Exactly in the definition we take $A = k[X]$, and then endomorphism will be determined by the image of X by the universal property of polynomial ring, then we can get the formal derivative of any polynomial

Definition 4.4. Let $\partial(X) = 1$, then for any $f = \sum_n a_n X^n \in k[X]$, we can define the formal derivative of f by

$$f'(X) := \partial(f) = \sum_n n a_n X^{n-1}$$

and a polynomial f is called **separable** if f is coprime with its formal derivative f' in $k[X]$.

Notice that $k[X]$ is a PID, so the coprime condition is equivalent to the existence of $r, s \in k[X]$ such that

$$1 = r(X)f(X) + s(X)f'(X)$$

we describe the separable polynomial without talking about roots of polynomial, and similarly we can give a definition of roots of a polynomial via arithmetic of polynomial:

Definition 4.5. Let $f \in k[X]$ and $a \in k$, then a is a **root** of f if there exists $k \in \mathbb{N}^*$ such that

$$(X - a)^k \mid f(X)$$

and the **order of the root** is defined as the largest integer m such that

$$(X - a)^m \mid f(X) \wedge (X - a)^{m+1} \nmid f(X)$$

in particular, a is a **simple root** if the order of the root is 1, and it is a multiple root if the order of the root is greater than 1.

Remark 4.5. Following this definition, $f(a) = 0$ if a is a root of f , the converse is not right in general, the ring should at least be an integral domain (**sure for field**) or the polynomial should be monic, the proof depends on the division of polynomial.

Proposition 4.7. For polynomial $f \in k[X]$, the following conditions are equivalent:

1. f is separable.
2. f has no multiple root in $\bar{k}$.
3. f and f' has no common root in an algebraic closure $\bar{k}$ of k .
4. f has exactly $\deg f$ distinct roots in $\bar{k}$.
5. $\bar{k}[X]/(f)$ is reduced and isomorphic to $\bar{k}^{\deg f}$.

Proof. (1) $\Rightarrow$ (2): If f has a multiple root a , then $(X - a)^2 \mid f(X)$, so $f(X) = (X - a)^2 g(X)$ for some $g \in k[X]$, then we can calculate the formal derivative of f :

$$f'(X) = 2(X - a)g(X) + (X - a)^2 g'(X)$$

hence we get $(X - a) \mid f(X)$ and $(X - a) \mid f'(X)$, which contradicts the coprime condition.

(2) $\Rightarrow$ (3): in $\bar{k}$, we can factorized f as linear terms, and the formal derivative of f can be calculated of the form

$$f'(X) = \sum_a \frac{f(X)}{X - a}$$

where a runs over all roots of f , we suppose that b is a common root of f' , then $f'(b) = f(X)/(X - b)|_{b=0} = 0$ implies that $X - b$ divides $f(X)/-b$, which gives a contradiction.

(3) $\Rightarrow$ (4): similar argument as above.

(4) $\Rightarrow$ (5): The isomorphism can be proved by the **chinese remainder theorem**, and the product of fields is reduced surely.

(5) $\Rightarrow$ (4): Notice that $\bar{k}[X]/(X-a)^2$ is not reduced with $\overline{X-a}^2 = 0$ in the quotient, and for any f we can factorized it to get a isomorphism

$$\bar{k}[X]/(f) \cong \prod_a \bar{k}[X]/(X-a)^{m_a}$$

with a distinct roots of f and m_a the multiplicity, so surely m_a should be all 1.

(4) $\Rightarrow$ (1): caculation. $\square$

actually we can characterize the irreducible polynomial if it is separable, it depends on the characteristic of the field.

Proposition 4.8. Let f be an irreducible polynomial in $k[X]$, then f is inseparable if and only if $f' = 0$, which shows that

- If $\text{Char}(k) = 0$, any irreducible polynomial is separable.
- If $\text{Char}(k) = p$, any irreducible polynomial is of the form $g(X^{p^r})$ with g a separable irreducible polynomial and $r \in \mathbb{N}$, and the representation is unique.

Proof. In $k[X]$ (a UFD), f and f' coprime means their gcd associates to 1, or equivalently the gcd is a unit. When f is irreducible, then f is a prime element in $k[X]$, so the gcd is either f or a unit. If f is inseparable, then the gcd is f , so $f|f'$ implies $f' = 0$ since $\deg f' < \deg f$. $\square$

Definition 4.6. Let L/K be a algebraic extension, then L/K is called a **separable extension** if for any $a \in L$, the minimal polynomial Π_a of a over K is separable.

Remark 4.6. **Any subextension of a separable extension is separable:** Let $K \subset E \subset L$ be a tower of algebraic extension, and L/K is separable. If $a \in E$, then $\Pi_{a,E}$ divides $\Pi_{a,L}$, and factor of separable polynomial is still separable (verify).

By proposition above, we can conclude that **any algebraic extension of a field with characteristic zero is separable**, in particular the extension of $\mathbb{Q}$, it is the core object in number theory. Hence, it is hopeless to find a example of inseparable extension of $\mathbb{Q}$ or $\mathbb{R}$, when we change our attention to field of non-zero characteristic, we can find a such example like beginning of this section.

Theorem 4.3. Let L/K be a finite extension, then the following conditions are equivalent:

1. L/K is separable.
2. $|\text{Hom}_K(L, \bar{K})| = [L : K]$
3. $\bar{K} \otimes_K L$ is reduced and is isomorphic to $\bar{K}^{[L:K]}$.

Proof. $\square$

Remark 4.7. When the extension is infinite, the condition 2 makes no sense, but it is still make sense to consider the condition 3. In the example of $L = \mathbb{F}_p(t^{1/p})$, we can find that the degree is p while the number of embedding is 1, so it is another way to show that L is an inseparable extension.

Corollary 4.9 (Primitive element theorem). Any finite separable extension is a simple extension, that is, for any finite separable extension L/K , there exists $a \in L$ such that $L = K(a)$, such a element a is called a **primitive element** of the extension.

Proof. □

Remark 4.8. If $L = K(a)$ a simple algebraic extension, we denote the roots of the minimal polynomial Π_a in $\bar{K}$ by R , then we can get a natural bijection:

$$\text{Hom}_K(L, \bar{K}) \rightarrow R, \quad f \mapsto f(a)$$

It is well-defined since embedding sends a to another root, and the map is always injective: For any $f, g \in \text{Hom}_K(L, \bar{K})$ with $f(a) = g(a)$, take any $x \in L = K(a)$, then there exists a polynomial $p \in K[X]$ such that $x = p(a)$, so we can calculate

$$f(x) = f(p(a)) = p(f(a)) = p(g(a)) = g(p(a)) = g(x)$$

which shows that $f = g$.

Hence the simple algebraic extension allows to identify the conjugations of the primitive element with the K -embeddings of the extension, and surely simple algebraic extension is not necessary separable, for example $L = \mathbb{F}_p(t^{1/p})$.

Discriminant

Proposition 4.10. Let L/K be a separable finite extension of degree n , then

$$- N_{L/K}(a) = \prod_{\sigma} \sigma(a)$$

$$- \text{Tr}_{L/K}(a) = \sum_{\sigma} \sigma(a)$$

$$- \chi_a(X) = \prod_{\sigma} (X - \sigma(a))$$

where $a \in L$ and $\sigma \in \text{Hom}_K(L, \bar{K})$ runs over all K -embeddings

Proof. We consider a relation of equivalence for $\Sigma = \text{Hom}_K(L, \bar{K})$ by

$$\sigma_1 \sim_a \sigma_2 \iff \sigma_1(a) = \sigma_2(a)$$

it holds for any finite extension, then we calculate the class of equivalence:

$$\begin{aligned} [\sigma] &= \{\tau \in \Sigma \mid \tau(a) = \sigma(a)\} \\ &= \{\tau \in \Sigma \mid \tau(p(a)) = \sigma(p(a)), \forall p \in K[X]\} \\ &= \{\tau \in \Sigma \mid \tau = \sigma \text{ on } K(a)\} \end{aligned}$$

which shows that $|\sigma| = |\text{Hom}_{K(a)}(L, \bar{K})|$, in particular, if L/K is separable, then $L/K(a)$ is also separable, so exactly $|\sigma| = [L : K(a)]$, it implies

$$\prod_{\sigma} (X - \sigma(a)) = \prod_{[\sigma] \in \Sigma / \sim_a} (X - \sigma(a))^{[L:K(a)]} = \Pi_a(X)^{[L:K(a)]} = \chi_a(X)$$

Here $\sigma(a)$ runs over all roots of Π_a since the restriction map

$$\text{res} : \text{Hom}_K(L, \bar{K}) \rightarrow \text{Hom}_K(K(a), \bar{K}), \quad \sigma \mapsto \sigma|_{K(a)}$$

is surjective. □

Definition 4.7. Let L/K be a finite extension of degree n

- The **trace pairing** of the extension is a symmetric K -bilinear form

$$Q_{L/K}(x, y) = \text{Tr}_{L/K}(xy)$$

- we define the **discriminant** of a family $\{e_1, \dots, e_n\}_{k \leq n}$ in L by

$$\text{Disc}(e_1, \dots, e_n) = \det(\text{Tr}_{L/K}(e_i e_j)) = \det(Q_{L/K}(e_i, e_j))$$

It is better to consider the discriminant in a separable extension, although we always use the **"free condition"** in the extension of $\mathbb{Q}$, the reason is as following:

Proposition 4.11. Let L/K be a finite extension, then the following conditions are equivalent:

1. L/K is separable.
2. $\text{Tr}_{L/K} \neq 0$.
3. The trace pairing $Q_{L/K}$ is non-degenerate.

Proof. □

Remark 4.9. Consider the inseparable example of $L = \mathbb{F}_p(t^{1/p})$, we can calculate $\text{Tr}_{L/K}(u^k) = 0$ with $u = t^{1/p}$ and $k = 0, 1, \dots, p-1$, which shows that the trace is identically zero, so the discriminant of any basis is zero, which provides no information about the extension.

Proposition 4.12. Let L/K be a finite separable extension of degree n , and $a_1, \dots, a_n$ be a basis of L/K , then the discriminant of the basis

$$\text{Disc}(a_1, \dots, a_n) = \det(\text{Tr}_{L/K}(\sigma_i(a_j)))^2$$

with $\sigma_i \in \text{Hom}_K(L, \bar{K})$ n distinct K -embeddings of L .

Proof. Review that for any two n order square matrices A, B , we have

$$(AB)_{i,j} = \sum_{k=1}^n A_{i,k} B_{k,j}$$

In particular

$$(A^T A)_{i,j} = \sum_{k=1}^n A_{k,i} A_{k,j}$$

so For any (i, j) -entry of trace matrix, applying proposition 4.10 we can calculate

$$\text{Tr}_{L/K}(a_i a_j) = \sum_{k=1}^n \sigma_k(a_i) \sigma_k(a_j) = \sum_{k=1}^n \sigma_k(a_i) \sigma_k(a_j) = \sum_{k=1}^n = (A^T A)_{i,j}$$

where $A = \text{Tr}_{L/K}(\sigma_i(a_j))_{1 \leq i, j \leq n}$. □

Corollary 4.13. Let L/K be a finite separable extension of degree n , then the discriminant of any basis of L/K is no-zero, in particular, if a is the primitive element such that $L = K(a)$ and $\{1, \dots, a^{n-1}\}$, then

$$\text{Disc}(1, a, \dots, a^{n-1}) = \prod_{i < j} (r_i - r_j)^2$$

where r_i are the distinct n roots of Π_a in $\bar{K}$.

Proof. Review that

$$(PAP^T)_{i,j} = \sum_{1 \leq k, l \leq n} P_{i,k} P_{j,l} A_{k,l}$$

For any two n order square matrices A, P . For any basis $\{e_1, \dots, e_n\}$, there exists a invertible matrix P such that $e_i = \sum_{j=1}^n P_{ij} a^{j-1}$, since $\{1, \dots, a^{n-1}\}$ is a basis, then

$$\begin{aligned} \text{Tr}_{L/K}(e_i e_j) &= \text{Tr}_{L/K} \left(\left(\sum_{k=1}^n P_{ik} a^{k-1} \right) \left(\sum_{l=1}^n P_{jl} a^{l-1} \right) \right) \\ &= \sum_{k,l=1}^n P_{ik} P_{jl} \text{Tr}_{L/K}(a^{k+l-2}) \\ &= \sum_{k,l=1}^n P_{ik} P_{jl} \text{Tr}_{L/K}(a^{k+l-2}) \end{aligned}$$

which shows that the trace matrix of the basis is $P^T A P$, where $A = (\text{Tr}_{L/K}(a^{i-1} a^{j-1}))_{i,j}$, which suffice to calculate that

$$\text{Disc}(e_1, \dots, e_n) = |\det(P)|^2 \text{Disc}(1, a, \dots, a^{n-1})$$

Hence just need to calculate the discriminant of the basis $\{1, \dots, a^{n-1}\}$, and prove that it is non-zero, which finishes the proof. By the above proposition, we can calculate

$$\text{Disc}(1, a, \dots, a^{n-1}) = \begin{vmatrix} 1 & \sigma_1(a) & \dots & \sigma_1(a)^{n-1} \\ 1 & \sigma_2(a) & \dots & \sigma_2(a)^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \sigma_n(a) & \dots & \sigma_n(a)^{n-1} \end{vmatrix}^2$$

where $\sigma_i(a)$ runs over all distinct roots since L/K is separable, hence we can conclude the result by the **Vandermonde determinant formula**. □

5 Algebraic approach to number theory

5.1 Number fields

To motivate the algebraic approach, we firstly consider a typical example in number theory, it will be the basis of the discussion.

- The finite extension of $K/\mathbb{Q}$ is called a **number field**, typically it is a separable extension since $\text{Char}(\mathbb{Q}) = 0$, so we can apply **primitive element** theorem to conclude that $K = \mathbb{Q}(a)$ for some $a \in K$.
- The element $a \in K$ is called an **algebraic integer** if there exists a monic polynomial $f(x) \in \mathbb{Z}[x]$ such that $f(a) = 0$. Notice that the following conditions are equivalent:
 - (a) a is an algebraic integer.
 - (b) minimal polynomial $\Pi_a(X) \in \mathbb{Z}[X]$.
 - (c) $\mathbb{Z}[a]$ is a finitely generated $\mathbb{Z}$ -module.

Proof. □

- Following the above definition, we can verify a similar result with the algebraic numbers: (a) sum of two algebraic integers is an algebraic integer, (b) product of two algebraic integers is an algebraic integer.

Proof. □

- It is trivial that algebraic integers are algebraic numbers, so similarly we can define

$$\tilde{\mathbb{Q}} = \{x \in \bar{\mathbb{Q}} : x \text{ is an algebraic integer}\}$$

It is a subring of algebraic closed field $\bar{\mathbb{Q}}$ or $\mathbb{C}$, and then we can define the ring of integers of number field K as

$$\mathcal{O}_K := \tilde{\mathbb{Q}} \cap K$$

equivalently, it is the set of all algebraic integers contained in K .

So we sketch the basic notation and definitions, the ring of integers of number field is special and important, since it is analogous to $\mathbb{Z}$ in $\mathbb{Q}$, and in particular number theory is the study of "integers", although here the integrality is not clear, so we will focus on it.

Theorem 5.1. Let $K/\mathbb{Q}$ be a number field with $\mathcal{O}_K$ its ring of integers, then

1. $(\mathcal{O}_K, +)$ is a free $\mathbb{Z}$ -module of rank $[K : \mathbb{Q}]$, hence it is a noetherian ring.
2. $\mathcal{O}_K$ is integrally closed in K (following the definition).
3. Any non-zero prime ideal in $\mathcal{O}_K$ is maximal, i.e. the Krull dimension of $\mathcal{O}_K$ is 1.

Proof. □

To study the arithmetic of "integers" we should understand the factorization of elements in $\mathcal{O}_K$, unfortunately, the ring of integers are not always UFD, typical example is $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ with $K = \mathbb{Q}(\sqrt{-5})$, in it we have

$$6 = 2 \times 3 = (1 + \sqrt{-5}) \times (1 - \sqrt{-5})$$

even more terrible, for the simplest example $K = \mathbb{Q}(\sqrt{d})$ (d is square-free), we have the following known results:

- When $d < 0$, $\mathcal{O}_K$ is a UFD if and only if d is one of the following integer:

$$-1, -2, -3, -7, -11, -19, -43, -67, -163$$

It is owe to the **Heegner-Baker-Stark theorem**, and the proof is non-trivial.

- When $d > 0$, it is conjectured by Gauss that there are infinitely many d such that $\mathcal{O}_K$ is a UFD, but it is still open (till 2026...).

Hence it is almost hopeless to consider the factorization of elements in $\mathcal{O}_K$. After dedekind and Kummer introduced the concept of ideal, a more general factorization is possible, in particular it holds for any number fields.

Theorem 5.2. For any number field K with ring of integers $\mathcal{O}_K$, any non-zero proper ideal $I \subsetneq \mathcal{O}_K$ can be factored into prime ideals, i.e. there exist prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ such that

$$I = \mathfrak{p}_1 \cdots \mathfrak{p}_n$$

and the factorization is unique up to permutation of the prime ideals.

For example we consider $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$, we have the following factorization of ideals:

$$(6) = \mathfrak{p}_2^2 \mathfrak{p}_3 \mathfrak{p}_3'$$

with

$$\mathfrak{p}_2 = (2, 1 + \sqrt{-5}), \quad \mathfrak{p}_3 = (3, 1 + \sqrt{-5}), \quad \mathfrak{p}_3' = (3, 1 - \sqrt{-5})$$

then we can find that the factorization of 6 is unique in the sense of ideals:

$$(2) = \mathfrak{p}_2^2, \quad (3) = \mathfrak{p}_3 \mathfrak{p}_3'$$

while

$$(1 + \sqrt{-5}) = \mathfrak{p}_2 \mathfrak{p}_3, \quad (1 - \sqrt{-5}) = \mathfrak{p}_2 \mathfrak{p}_3'$$

To prove the theorem, some preparation should be done since the result is essentially not trivial, firstly we should consider the division of ideals:

Definition 5.1. Let $\mathfrak{a}$ and $\mathfrak{b}$ be two non-zero ideals in $\mathcal{O}_K$, we define

$$\mathfrak{a} | \mathfrak{b} \iff \mathfrak{b} \subset \mathfrak{a}$$

This definition can be done for any commutative ring, but the properties of rings of integers ensures the following results, which is the key to the proof.

Lemma 5.1. $\mathfrak{p}$ is a prime ideal of $\mathcal{O}_K$ if and only if for any ideals $\mathfrak{a}$ and $\mathfrak{b}$ in $\mathcal{O}_K$, we have $\mathfrak{p}|\mathfrak{a}\mathfrak{b} \implies \mathfrak{p}|\mathfrak{a}$ or $\mathfrak{p}|\mathfrak{b}$.

Proof. □

Lemma 5.2. For any non-zero ideals $\mathfrak{a} \in \mathcal{O}_K$, there exists non-zero prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ such that $\mathfrak{p}_1 \cdots \mathfrak{p}_n \subset \mathfrak{a}$.

Proof. □

Lemma 5.3. For any prime ideal $\mathfrak{p}$ in $\mathcal{O}_K$, we define its inverse ideal as

$$\mathfrak{p}^{-1} = \{x \in K : x\mathfrak{p} \subset \mathcal{O}_K\}$$

then for any non-zero ideal $\mathfrak{a}$ in $\mathcal{O}_K$, we have $\mathfrak{a} \subsetneq \mathfrak{a}\mathfrak{p}^{-1}$. In particular, $\mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}_K$.

Proof. □

With the two lemma we can finish the proof:

Proof of theorem 5.2. For the existence, we set a set S of ideals which can not be factorized into prime ideals, if S is empty then we are done, **hence we assume S is non-empty**. We consider the partial ordered set $(S, \subset)$, then the noetherian condition ensures the existence of maximal element, denoted by $\mathfrak{a}$.

$\mathfrak{a}$ can not be prime (i.e. maximal), otherwise it is already factorized, hence there exists a maximal ideal (i.e prime) such that $\mathfrak{a} \subsetneq \mathfrak{p}$, which implies

$$\mathfrak{a} \subsetneq \mathfrak{a}\mathfrak{p}^{-1} \subsetneq \mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}_K$$

the chain holds true by lemma 5.3, hence we construct an element $\mathfrak{a}\mathfrak{p}^{-1} \notin S$, which allows us to factorize

$$\mathfrak{a}\mathfrak{p}^{-1} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$$

then $\mathfrak{a} = \mathfrak{a}\mathcal{O}_K = \mathfrak{a}\mathfrak{p}^{-1}\mathfrak{p}$ shows that the maximal element can be factorized, which is a contradiction.

For the uniqueness, we assume that there are two factorization for an ideal $\mathfrak{a}$

$$\mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_n = \mathfrak{q}_1 \cdots \mathfrak{q}_m$$

then $\mathfrak{p}_1|\mathfrak{a}$ shows that there exists i such that $\mathfrak{p}_1|\mathfrak{q}_i$, prime ideals are maximal so $\mathfrak{p}_1 = \mathfrak{q}_i$ exactly, and we can cancel then by multiplying $\mathfrak{p}^{-1}$, then we can repeat the process to conclude that $n = m$ and $\mathfrak{p}_i = \mathfrak{q}_{\sigma(i)}$ for any permutation $\sigma \in S_n$. □

6 Analytic approach to number theory

6.1 Character

Definition 6.1. Let G be a finite abelian group, a **character** of G is a group homomorphism $\chi : G \rightarrow \mathbb{C}^\times$, and we denote the set of all characters of G as $\hat{G}$, it forms a group under function multiplication

$$\forall g \in G : (\chi_1 \cdot \chi_2)(g) := \chi_1(g)\chi_2(g)$$

And we call $\hat{G}$ the **Character group** of G .

Here are some quick observations:

- **image of character should be in $\mathbb{S}^1$:** any elements $a \in G$ is of finite order, and homomorphism of groups makes $\chi(e) = 1$, hence $\chi(a)^n = 1$ with $n = \text{ord}(a)$.
- **Inverse of character is the complex conjugate:** $|\chi(n)| = 1$, which allows us to define the inverse

$$\chi^{-1}(n) = 1/\chi(n) = \bar{\chi}(n)$$

- In particular, $\chi(-1) = \pm 1$ since $\chi(-1)^2 = \chi(1) = 1$, which allows us to classify the character into two types: **even** character with $\chi(-1) = 1$ and **odd** character with $\chi(-1) = -1$.

Remark 6.1. We consider the category of all abelian groups, denoted by **Ab**, the character group can be generalized in the category by a contravariant functor $(\hat{-}) : \mathbf{Ab} \rightarrow \mathbf{Ab}^{op}$: For the object $G \in \mathbf{Ab}$, we have $\hat{G} = \text{Hom}(G, \mathbb{C}^\times)$; For the morphism $f : G \rightarrow H$, it induces a morphism between dual groups $\hat{f} : \hat{H} \rightarrow \hat{G}$ by $\chi \mapsto \chi \circ f$, i.e.

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ & \searrow \hat{f} & \downarrow \chi \\ & & \mathbb{C}^* \end{array}$$

However, the functor is not an equivalent since it is not full, that is the reason why we only consider the character group of finite abelian groups.

Remark 6.2. Another view of character is that the character is naturally a one-dimensional complex representation of G since $\mathbb{C}^* \cong \text{GL}_1(\mathbb{C})$, so any character as a representation is irreducible.

Proposition 6.1. For any finite abelian group G , we have $\hat{\hat{G}} \cong G$

Proof. For any finite abelian group G , there exists a decomposition of cyclic groups $G \cong \mathbb{Z}/n_1 \oplus \cdots \oplus \mathbb{Z}/n_r$, the universal property of direct sum allows $\widehat{G \oplus H} \cong \hat{G} \times \hat{H}$ hence we have

$$\hat{G} \cong \widehat{\mathbb{Z}/n_1} \oplus \cdots \oplus \widehat{\mathbb{Z}/n_r}$$

hence it is enough to prove the isomorphism for any cyclic group $\mathbb{Z}/n$. □

Here is a useful lemma, which appears in many proofs, from the view of category, it shows that the induced morphism of an injection is a surjection.

Lemma 6.2. Let G be a finite abelian group and H be its subgroup, then for any character $\chi \in \hat{H}$, there exists exactly $[G : H]$ characters $\hat{G}$ that extend χ to G .

Proof. Consider restriction map $r : \hat{G} \rightarrow \hat{H}$ defined by $\chi \mapsto \chi|_H$, it is a group homomorphism, and we will show that r is surjective, then we consider the equivalence relation on $\hat{G}$ by

$$\chi_1 \sim \chi_2 \iff \chi_1|_H = \chi_2|_H$$

the relation gives that the class of χ is exactly the set of characters that extend χ to G , and the relation is actually same as the relation defined by cosets of $\bar{H}$, then we can deduce the number by isomorphism of dual groups:

$$\#[\chi] = [\hat{G} : \hat{H}] = [G : H]$$

Surjective: We take $g_1 \notin H$ and construct a subgroup of G by

$$G_1 = \langle H, g_1 \rangle = \{g_1^k h \mid k \in \mathbb{N}, h \in H\}$$

then we can extend the character χ by defining $\chi_1(g_1^k h) = \chi(h)$, i.e. we set $\chi_1(g_1) = 1$ and allow $\chi_1(ab) = \chi_1(a)\chi_1(b)$, then we can verify that $\chi_1 \in \widehat{G_1}$ and $\chi_1|_H = \chi$. Similarly we can construct a chain of subgroups by $G_k = \langle G_{k-1}, g_k \rangle$ with some $g_k \notin G_{k-1}$, so we have

$$G_0 = H \subsetneq G_1 \subsetneq \cdots \subsetneq G_n = G$$

the chain will stop since G is finite (or it is finitely generated $\mathbb{Z}$ -module, so we can use noetherian condition). $\square$

Remark 6.3. Notice that $\mathbb{C}^*$ is a divisible abelian group, then for any short sequence of abelian groups

$$0 \rightarrow H \rightarrow G \rightarrow G/H \rightarrow 0$$

we have a conversion of exact sequence

$$0 \rightarrow \widehat{G/H} \rightarrow \hat{G} \rightarrow \hat{H} \rightarrow 0$$

we can talk more if we restrict our view on finite abelian groups.

Theorem 6.3. Let G be a finite abelian group, then any quotient group of G can be seen as a subgroup of G , furthermore we have

$$G \cong H \oplus G/H$$

Proof. We set $H^\perp = \{\chi \in \hat{G} \mid \chi(h) = 1, \forall h \in H\}$ as the complement of $\hat{H}$ in $\hat{G}$, it is natural in the sense of $\mathbb{Z}$ -module, then we can prove that $H^\perp \cong \widehat{G/H}$. Notice that $H^\perp$ is exactly a subgroup of $\hat{G}$, and $H^\perp \cap \hat{H} = \{1_H\}$, so we have a direct sum

$$\hat{G} = \hat{H} \oplus H^\perp \cong \hat{H} \oplus \widehat{G/H}$$

by the isomorphism of dual groups, we can deduce the result. $\square$

Proposition 6.4 (orthogonality). Let G be a finite abelian group,

1. Fix $\chi \in \bar{G}$, then

$$\sum_{g \in G} \chi(g) = \begin{cases} |G| & \chi = 1 \\ 0 & \chi \neq 1 \end{cases}$$

2. Fix $g \in G$, then

$$\sum_{\chi \in \hat{G}} \chi(g) = \begin{cases} |G| & g = e \\ 0 & g \neq e \end{cases}$$

Proof. For the first case, if $\chi = 1$ then it is trivial, otherwise we can take $h \in G$ such that $\chi(h) \neq 1$, and the translation $g \mapsto h^{-1}g$ is a bijection of G , so we have

$$\sum_{g \in G} \chi(g) = \sum_{g \in G} \chi(h)\chi(h^{-1}g) = \chi(h) \sum_{g \in G} \chi(g)$$

then we can deduce the result. For the second case, we suppose that $g \neq e$, then there exists a character ψ such that $\psi(g) \neq 1$, and the translation $\chi \mapsto \psi^{-1} \cdot \chi$ is a bijection of $\hat{G}$, so we have

$$\sum_{\chi \in \hat{G}} \chi(g) = \psi(g) \sum_{\chi \in \hat{G}} (\psi^{-1} \cdot \chi)(g) = \psi(g) \sum_{\chi \in \hat{G}} \chi(g)$$

then we can deduce the result. $\square$

A special result of orthogonality is about the structure of the character, we set **FinAb** to be the category of finite abelian groups, as we have mentioned before, there exists a contravariant functor inside the category, here we set $D : \mathbf{FinAb} \rightarrow \mathbf{FinAb}^{op}$, the following result shows that the functor is actually an equivalence with $D^2 = Id$.

Theorem 6.1. Let G be a finite abelian group, there exists a natural isomorphism $\phi : G \rightarrow \hat{\hat{G}}$ defined by $g \mapsto \phi_g$ with $\phi_g(\chi) = \chi(g)$.

Proof. It is easy to verify that the map is a morphism, so it is enough to show injection since it is finite, and we consider the kernel.

$$\begin{aligned} g \in \ker \phi &\iff \phi_g(\chi) = 1, \forall \chi \in \hat{G} \\ &\iff \chi(g) = 1, \forall \chi \in \hat{G} \\ &\iff \sum_{\chi \in \hat{G}} \chi(g) = |\hat{G}| = |G| \\ &\iff g = e \end{aligned}$$

$\square$

Remark 6.4. Notice that the map is «natural» since it does not depend on the choice of groups, and we can verify the following diagram commutes for any morphism $f : G \rightarrow H$

$$\begin{array}{ccc} G & \xrightarrow{\phi_G} & \hat{\hat{G}} \\ f \downarrow & & \downarrow D^2(f) \\ H & \xrightarrow{\phi_H} & \hat{\hat{H}} \end{array}$$

which ensures that the functor D is an equivalence

6.1.1 Legendre Symbol

A character of a finite abelian group is called **real character** if its image is contained in $\{\pm 1\}$, this part is to introduce a typical example of real character.

Historically, a classical diophantine problem is to solve

$$X^2 = b \pmod{p}$$

with p a prime, and b is a given non-zero integer, equivalently, solve the quadratic equation in the finite field $\mathbb{F}_p$. For example, $X^2 - 2 = 0$ has no solution in $\mathbb{F}_5$ while it has solution in $\mathbb{F}_3$.

Proposition 6.5. Let p be an odd prime, then $\varphi : \mathbb{F}_p^* \rightarrow \mathbb{F}_p^*$ with $x \mapsto x^{\frac{p-1}{2}}$ defines a homomorphism of groups, with kernel S_p the set of all squares in $\mathbb{F}_p^*$, and the image is $\{1, -1\}$.

Proof. p is prime so $\frac{p-1}{2}$ is an integer, which is enough to show the map is a homomorphism. Notice that $S_p \subset \ker \varphi$, it is enough to show that $|S_p| = |\ker \varphi| = \frac{p-1}{2}$. After that, the cardinality of the image is 2, and $\varphi(x)$ is exactly the roots of $X^2 - 1 = 0$, which implies the image is $\{1, -1\}$.

S_p is the image of morphism $f(x) = x^2$ inside $\mathbb{F}_p^*$, the kernel of f is exactly the solutions of $X^2 - 1 = 0$ in $\mathbb{F}_p$, they are exactly $-1, 1$ since there are at most two distinct solutions, hence $|S_p| = |\mathbb{F}_p^* / \ker f| = \frac{p-1}{2}$.

$\varphi(x) = 1$ means that x is a root of $X^{\frac{p-1}{2}} - 1 = 0$ in $\mathbb{F}_p$, which has at most $\frac{p-1}{2}$ distinct solutions, hence $|\ker \varphi| \leq \frac{p-1}{2} = |S_p|$, so immediately $\ker \varphi = S_p$. $\square$

Notice that the image of homomorphism is again a group, and the group $\{-1, 1\}$ in $\mathbb{F}_p^*$ is isomorphic to $\mathbb{Z}/2\mathbb{Z}$, so it has no problem to identify $\{-1, 1\}$ in $\mathbb{F}_p$ as $\{-1, 1\}$ in $\mathbb{C}^*$, and we define the **Legendre symbol** as a character of $\mathbb{F}_p^*$ by

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & a \in S_p \\ -1 & a \notin S_p \end{cases}$$

6.1.2 Dirichlet Character

Definition 6.2. Let $N \in \mathbb{N}^*$, a **Dirichlet character** modulo N is a character of the multiplicative group $(\mathbb{Z}/N\mathbb{Z})^\times$, i.e.

$$\chi : (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$$

Here are some quick descriptions of this type of character:

- The order of $(\mathbb{Z}/N\mathbb{Z})^\times$ is $\varphi(N)$ with φ the Euler totient function, so the orthogonality of characters can be written as

$$\sum_{a \in (\mathbb{Z}/N\mathbb{Z})^\times} \chi(a) = \begin{cases} \varphi(N) & \chi = 1 \\ 0 & \chi \neq 1 \end{cases} \quad \sum_{\chi \in \widehat{(\mathbb{Z}/N\mathbb{Z})^\times}} \chi(g) = \begin{cases} \varphi(N) & g = 1 \\ 0 & g \neq 1 \end{cases}$$

- Let χ be a Dirichlet character modulo N , then we can extend χ to a function on $\mathbb{Z}$ by

$$\tilde{\chi}(a) = \begin{cases} \chi(\bar{a}) & \gcd(a, N) = 1 \\ 0 & \gcd(a, N) \neq 1 \end{cases}$$

It defines an arithmetic function that is **completely multiplicative** and **N -periodic**. Each character can be extended like that, so we sometimes do not distinguish the character and the arithmetic function.

Theorem 6.6. Let $f : \mathbb{Z} \rightarrow \mathbb{C}$ be an arithmetic function is a Dirichlet character modulo N if it satisfies the following conditions:

- (1) completely multiplicative.
- (2) N -periodic for some $N \in \mathbb{N}^*$.
- (3) $|f(n)| = 1$ if and only if $\gcd(n, N) = 1$.

Proof.

□

In particular, we noticed that as a real character, the Legendre symbol is a Dirichlet character modulo p .